

2018 Level I Mock Exam (B) AM

The morning session of the 2018 Level I Chartered Financial Analyst® Mock Examination has 120 questions. To best simulate the exam day experience, candidates are advised to allocate an average of one and a half minutes per question for a total of 180 minutes (3 hours) for this session of the exam.

Questions	Topic	Minutes
1–18	Ethical and Professional Standards	27
19–33	Quant	22.5
34–45	Econ	18
46–69	Financial Reporting and Analysis	36
70–78	Corporate Finance	13.5
79–86	Portfolio Management	12
87–98	Equity	18
99–110	Fixed Income	18
111–115	Derivatives	7.5
116–120	Alternative Investments	7.5
Total:		180

By accessing this mock exam, you agree to the following terms of use: This mock exam is provided to currently registered CFA candidates. Candidates may view and print the exam for personal exam preparation only. The following activities are strictly prohibited and may result in disciplinary and/or legal action: accessing or permitting access by anyone other than currently-registered CFA candidates; copying, posting to any website, emailing, distributing and/or reprinting the mock exam for any purpose

© 2017 CFA Institute. All rights reserved.

2018 LEVEL I MOCK EXAM (B) AM

- 1 Justin Blake, CFA, a retired portfolio manager, owns 20,000 shares of a small public company that he would like to sell because he is worried about the company's prospects. He posts messages on several internet bulletin boards. The messages read, "This stock is going up once the pending patents are released, so now is the time to buy. The stock is a buy at anything below \$3. I have done some close research on these guys." According to the *Standards of Practice Handbook*, Blake *most likely* violated the Standard or Standards associated with:
- A Integrity of Capital Markets and Conflicts of Interest.
 - B Integrity of Capital Markets, but not Conflicts of Interest.
 - C Neither Integrity of Capital Markets nor Conflicts of Interest.

B is correct because Blake violated the Integrity of Capital Markets by engaging in a practice that is likely to artificially inflate trading volume [Standard II(B)].

A is incorrect because the Conflicts of Interests [Standard VI(A)] requires disclosure of conflicts to clients, prospective clients, and employers, which is not the case in this example.

C is incorrect because Blake violated the Integrity of Capital Markets Standard.

Guidance for Standards I–VII

LOS a

Standard II(B)–Market Manipulation, Standard VI(A)–Disclosure of Conflicts

- 2 According to the CFA Institute Code of Ethics and Standards of Professional Conduct, trading on material nonpublic information is *least likely* to be prevented by establishing:
- A firewalls.
 - B selective disclosure.
 - C personal trading limitations.

B is correct as selective disclosure occurs when companies discriminate in making material nonpublic information public. Corporations that disclose information on a limited basis create the potential for insider-trading violations. Standard II(A).

A is incorrect as an information barrier commonly referred to as a "firewall" is a widely used approach to preventing the communication of material nonpublic information within firms.

C is incorrect as limitations on personal trading by employees is one of the recommended procedures for compliance to prevent employees from trading on material nonpublic information.

Guidance for Standards I–VII

LOS c

Standard II(A)–Material Nonpublic Information

- 3 Jefferson Piedmont, CFA, a portfolio manager for Park Investments, plans to manage the portfolios of several family members in exchange for a percentage of each portfolio's profits. As his family members have extensive portfolios requiring substantial attention, they have requested that Piedmont provide the services outside his employment with Park. Piedmont notifies his employer in writing of his prospective outside employment. Two weeks later, Piedmont begins managing the family members' portfolios. By managing these portfolios, which of the following CFA Institute Standards of Professional Conduct has Piedmont violated?
- A Conflicts of Interest
 - B Additional Compensation
 - C Both Additional Compensation and Conflicts of Interest

C is correct because members should disclose all potential conflicts of interest, the substantial time involved in managing family accounts, and when engaging in independent practice for compensation should not render services until receiving written consent from all parties [Standard IV(B), Standard VI(A)].

A is incorrect because both standards have been violated.

B is incorrect because both standards have been violated.

Guidance for Standards I–VII

LOS a

Standard IV(B)–Additional Compensation Arrangements, Standard VI(A)–Disclosure of Conflicts

- 4 Noor Mawar, CFA, manages a trust fund with the beneficiary being an orphaned 18-year-old student. The investment policy dictates that trust assets are expected to provide the student with a stable low-risk source of income until she reaches the age of 30 years. Based on information from an Internet blog, the student asks Mawar to invest in a new business venture that she expects will provide high returns over the next five years. Mawar ignores the request, instead securing conservative investments to provide sufficient income. Did Mawar *most likely* violate the CFA Institute Code of Ethics and Standards of Professional Conduct?
- A Yes.
 - B No, because the client's objectives were met.
 - C No, because the investment time frame does not match the investment horizon.

B is correct because the client is the trust/trustees, not the beneficiary. Mawar followed Standard III(C) –Suitability by managing the trust assets in a way that would likely result in a stable source of income while keeping the risk profile low, thereby complying with the investment objectives of the trust.

A is incorrect because Mawar did not violate any Standard as she managed trust assets considering the suitability for the client, not the beneficiary.

C is incorrect because the client is the trust/trustees, not the beneficiary. Therefore the time horizon of the investment is not relevant.

Guidance for Standards I–VII
LOS b
Standard III(C)–Suitability

- 5 After a firm presents a minimum required number of years of GIPS-compliant performance, the firm must present an additional year of performance each year, building up to a minimum of:
- A 10 years of GIPS-compliant performance.
 - B 5 years of GIPS-compliant performance.
 - C 15 years of GIPS-compliant performance.



A is correct. After a firm presents a minimum of five years of GIPS-compliant performance, the firm must present an additional year of performance each year, building up to a minimum of 10 years of GIPS-compliant performance.

B is incorrect. The initial period is five years of GIPS-compliant performance, building up to a minimum of 10 years of GIPS-compliant performance.

C is incorrect. An additional year of performance each year is added building up to a minimum of 10 years of GIPS-compliant performance.

The GIPS Standards
LOS b

- 6 In the event of a discrepancy between the official GIPS standards and the local language translation, the official governing language is:
- A English.
 - B the language of the local country.
 - C the language of a neutral country.



A is correct. Although the GIPS standards may be translated into many languages, if a discrepancy arises, the English version of the GIPS standards is the official governing version.

B is incorrect. The English version of the GIPS standards is the official governing version, not the language of the local country.

C is incorrect. The English version of the GIPS standards is the official governing version, not the language of a neutral third country.

The GIPS Standards
LOS c

- 7 Disclosure of confidential CFA exam information will *most likely* be detected by the Professional Conduct staff through:
- A monitoring online and social media.
 - B analysis of Proctor Reports.
 - C annual Professional Conduct Statements.



A is correct. Professional Conduct inquiries come from a number of sources including the monitoring of online and social media to detect disclosure of confidential exam information.

B is incorrect. Candidate conduct is monitored by exam proctors who complete reports on candidates suspected to have violated testing rules during the exam and at the exam center.

C is incorrect. Members and candidates must self-disclose on the annual Professional Conduct Statement all matters that question their professional conduct, such as involvement in civil litigation or a criminal investigation or being the subject of a written complaint. Disclosure of confidential exam information will not be found on the annual statement.

Code of Ethics and Standards of Professional Conduct
LOS a

- 8 When making performance presentations to prospective clients, a GIPS compliant firm should *least likely* do which of the following?
- A Selectively report its top fund performance
 - B Include the fund performance of former clients
 - C Report performance history for all market cycles under review



A is correct. When a firm complies with GIPS standards it cannot selectively choose its top fund performances while excluding weaker performing funds. It must include all fee-paying discretionary funds managed to a similar investment mandate, objective, or strategy.

B is incorrect because GIPS standards require the performance of former clients to be included to avoid survivorship bias.

C is incorrect because GIPS standards require the fund performance to reflect results across all market cycles for the periods under review. Fund managers are not allowed to select a time period during which the mandate produced superior results while leaving out other time period that underperformed (varying time periods).

Introduction to the Global Investment Performance Standards (GIPS)
LOS a
Section I

- 9 A central bank fines a commercial bank it supervises for not following statutory regulations regarding non-performing loan provisions on three large loans as a result of the bank's loan provisioning policy. Louis Marie Buffet, CFA, sits on the Board of Directors of the commercial bank as a non-executive director, representing minority shareholders. He also chairs the internal audit committee of the bank that determines the loan provisioning policy of the bank. Mercy Gatabaki, CFA, is the bank's external auditor and follows international auditing standards whereby she tests the loan portfolio by randomly selecting loans to check for compliance in all aspects of central bank regulations. Which charterholder is *most likely* in violation of the Code and Standard?
- A Both.
 - B Buffet.
 - C Gatabaki.



B is correct because Buffet sat on the audit committee that determined the bank's provisioning policies that were contrary to the statutory regulations of the central bank. As a result, he most likely violated Standard I–Professionalism by not abiding with regulations of a regulatory body. Gatabaki did not violate Standard I–Professionalism as it is not apparent she knowingly facilitated the incorrect provisioning policy.

A is incorrect because only Buffet is most likely to have violated Standard I–Professionalism by not following central bank regulations.

C is incorrect because Gatabaki most likely did not violate Standard I–Professionalism as it is not apparent she knowingly facilitated the incorrect provisioning policy or followed it. Gatabaki randomly selects loans for her audit and given the relatively small number of loans in violation, it is likely she did not come across the files which were treated in a manner by the bank that was contrary to the central bank regulations.

Guidance for Standards I–VII

LOS a

Standard I(A)–Knowledge of the Law

- 10** Diana Fairbanks, CFA, is married to an auditor who is employed at a large accounting firm. When her husband mentions that a computer firm he audits will receive a qualified opinion she thinks nothing of it. Later that week when she reviews a new client account she notices that there are substantial holdings of this computer firm. When she does a thorough internet search for news on the company, she does not find anything about its most recent audit or any other adverse information. Which of the following actions concerning the computer stock should Fairbanks *most likely* take to avoid violating the CFA Institute Standards of Professional Conduct?

- A** Take no investment action.
- B** Complete a thorough and diligent analysis of the company and then sell the stock.
- C** Sell the stock immediately as she has a reasonable basis for taking this investment action.



A is correct as the information concerning the qualified opinion is nonpublic and if it is material she would be in violation of Standard II(A) if she took investment action based on the information. She should also make reasonable efforts to achieve public dissemination of the information.

B is incorrect because she should make reasonable efforts to achieve public dissemination of the information prior to selling the stock.

C is incorrect because she should make reasonable efforts to achieve public dissemination of the information prior to selling the stock.

Guidance for Standards I–VII

LOS c

Standard II(A)–Material Nonpublic Information

- 11** Sherry Buckner, CFA, manages equity accounts for government entities whose portfolios are classified as being conservative and risk averse. Since the objective of her clients is to maximize returns with the lowest possible risk, Buckner considers adding to their holdings a new, thinly traded, leveraged derivative

product that she believes has the potential for high returns. To make her investment decision, Buckner relies upon comprehensive research from an investment bank with a solid reputation for top quality research. After her review of that research, Buckner positions her accounts so each has a 10% allocation to the derivative product. Did Buckner *most likely* violate any CFA Institute Standards of Professional Conduct by purchasing the derivative product for her clients?

- A No.
- B Yes, related to Suitability.
- C Yes, related to Loyalty, Prudence, and Care.

B is correct as Buckner is in violation of Standard III(C) since she did not consider issues such as the limited liquidity or any potential leverage of this new product when she invested a substantial percentage of her clients' portfolios in these instruments.

A is incorrect because Buckner violated the suitability Standard.

C is incorrect because Buckner relied upon comprehensive research from the investment bank.

Guidance for Standards I–VII
LOS b
Standard III(C)–Suitability

- 12 Which of the following is *least likely* part of the CFA Institute Standards of Professional Conduct, Standard V(B)–Communication with Clients and Prospective Clients? Members and candidates must:
- A make reasonable efforts to ensure that when communicating investment performance information it is fair, accurate, and complete.
 - B disclose to clients and prospective clients significant limitations and risks associated with the investment process.
 - C distinguish between fact and opinion in the presentation of investment analysis and recommendations.

A is correct. The statement, "When communicating investment performance information, Members and Candidates must make reasonable efforts to ensure that it is fair, accurate, and complete." can be found in The CFA Institute Standards of Professional Conduct, Standard III–Duties to Clients (D) Performance Presentation. It is not part of Standard V–Investment Analysis, Recommendations, and Actions (B) Communication with Clients and Prospective Clients.

B is incorrect. The statement, "Members and Candidates must disclose to clients and prospective clients significant limitations and risks associated with the investment process" can be found in The CFA Institute Standards of Professional Conduct, Standard V–Investment Analysis, Recommendations, and Actions (B) Communication with Clients and Prospective Clients.

C is incorrect. The statement, “Members and Candidates must distinguish between fact and opinion in the presentation of investment analysis and recommendations” can be found in The CFA Institute Standards of Professional Conduct, Standard V–Investment Analysis, Recommendations, and Actions (B) Communication with Clients and Prospective Clients.

Code of Ethics and Standards of Professional Conduct
LOS b

- 13** Jack Steyn, CFA, recently became the head of the trading desk at a large investment management firm that specializes in domestic equities. While reviewing the firm’s trading operations he notices that clients give discretion to the manager to select brokers on the basis of their overall services to the management firm. Despite the client directive, Steyn would *most likely* violate Standard III(A)–Loyalty, Prudence, and Care if he pays soft commissions for which of the following services from the brokers?
- A** Equity research reports
 - B** Investment conference attendance
 - C** Database services for offshore investments



C is correct because Standard III(A)–Loyalty, Prudence, and Care stipulates that the client owns the brokerage. Therefore, members and candidates are required to only use client brokerage to the benefit of the clients (soft commissions policy). As the firm specializes in domestic equity, an offshore investment database service would not benefit clients.

A is incorrect because it is likely that equity research reports would benefit all clients.

B is incorrect because it is likely that attendance at an investment conference could lead to ideas and subsequent investment actions that would benefit all clients.

Guidance for Standards I–VII
LOS b

- 14** Henrietta Huerta, CFA, writes a weekly investment newsletter to market her services and obtain new asset management clients. A third party distributes the free newsletter on her behalf to those individuals on its mailing list. As a result, it is widely read by thousands of individual investors. The newsletter recommendations reflect most of Huerta’s investment actions. After completing further research on East-West Coffee Roasters, Huerta decides to change her initial buy recommendation to a sell. To avoid violating the CFA Institute Standards of Professional Conduct it would be *most* appropriate for Huerta to distribute the new investment recommendation to:
- A** newsletter recipients first.
 - B** asset management clients first.
 - C** newsletter recipients and asset management clients simultaneously.



B is correct because according to Standard III(A)–Loyalty, Prudence, and Care, members and candidates must place their clients’ interests first before their own interests. The temptation may be to release the changed recommendation to newsletter recipients

simultaneously with or even before the asset management clients to try to obtain new clients. However, to avoid violating Standard III(A)–Loyalty, Prudence, and Care, Huerta must ensure that any change in an investment recommendation is first distributed to her asset management clients before any newsletter recipients, who are not necessarily clients (that is, they receive the newsletter for free from a third party distribution list).

A is incorrect because according to Standard III(A)–Loyalty, Prudence, and Care members and candidates when making investment recommendations (or changing investment recommendations) must give priority to clients; i.e., asset management clients rather than to non-clients; i.e., newspaper recipients who receive the newsletter for free from a third party distribution list.

C is incorrect because according to Standard III(A)–Loyalty, Prudence, and Care, members and candidates when making investment recommendations (or changing investment recommendations) must give priority to clients, i.e., asset management clients rather than to non-clients, i.e., newspaper recipients who receive the newsletter for free from a third party distribution list.

Guidance for Standards I–VII
LOS c

- 15 Wouter Duyck, CFA, is the sole proprietor of an investment advisory firm serving several hundred middle class retail clients. Duyck claims to be different from his competitors because he conducts research himself. He discloses that to simplify the management of all these accounts he has created a recommended list of stocks, from which he selects investments for all of his clients based on their suitability. Duyck's recommended list of stocks is obtained from his primary broker, who has completed due diligence on each stock. Duyck's recommended list *least likely* violates which of the following CFA Institute Standards of Professional Conduct?
- A Fair Dealing.
 - B Misrepresentation.
 - C Diligence and Reasonable Basis.



A is correct because Standard III(B)–Fair Dealing concerns the fair treatment of clients when making investment recommendations or taking investment action, but there is no indication that the advisor has discriminated against any clients with regard to his recommendations as he invests all clients in the same universe of stocks. The advisor has violated Standard I(C)–Misrepresentation with his research, which is not independently created and instead relies upon information provided by his broker. This is contrary to the advisor telling clients he does his own independent investment research. In addition, the advisor has violated Standard V(A)–Diligence and Reasonable Basis, as he has not made reasonable and diligent efforts to determine if the third party's research is sound.

B is incorrect, as the advisor has violated Standard I(C)–Misrepresentation with his research, which is not independently created and instead relies upon information provided by his broker.

C is incorrect, as the advisor has violated Standard V(A)–Diligence and Reasonable Basis as he does not have a reasonable basis for making his investment recommendations and relies solely on his broker’s research to create his list of stock investments. This is directly contrary to telling clients that he does his own independent investment research.

Guidance for Standards I–VII

LOS b

Standard I(C)–Misrepresentation, Standard III(B)–Fair Dealing, Standard V(A)–Diligence and Reasonable Basis

- 16** Heidi Halvorson, CFA, is the Chief Investment Officer for Tukwila Investors, an asset management firm specializing in fixed-income investments. Tukwila is in danger of losing one of its largest clients, Quinault Jewelers, which accounts for nearly one third of its revenues. Quinault recently told Halvorson that Tukwila would be fired unless the performance of Quinault’s portfolio improves significantly. Shortly after this conversation, Halvorson purchases two corporate bonds she believes are suitable for any of her clients based upon third party research from a reliable and diligent source. Immediately after the purchase, one bond increases significantly in price while the other bond declines significantly. At the end of the day, Halvorson allocates the profitable bond trade to Quinault and the other bond to two of her largest institutional accounts. Halvorson *most likely* violated the CFA Institute Standards of Professional with regards to:

- A** client suitability.
- B** trade allocations.
- C** third party research.



B is correct because the investment officer failed to deal fairly for her clients by allocating profitable trades to a favored client at the expense of others, a violation of Standard III(B)–Fair Dealing. The standard requires members and candidates to treat all clients fairly when taking investment action. Tukwila should have a systematic approach to allocating trades, such as pro rata, before or at the time of trade execution or as soon as possible after trades are executed.

A is incorrect because the analyst believes the bonds are suitable for any of her clients and has not violated Standard III(C)–Suitability.

C is incorrect because the analyst does have a reasonable or adequate basis for her investment decision, because it is based upon reliable third party research, and has not violated Standard V(A)–Diligence and Reasonable Basis.

Guidance for Standards I–VII

LOS b

Standard III(B)–Fair Dealing, Standard III(C)–Suitability, Standard V(A)–Diligence and Reasonable Basis

- 17** Alan Quanta, CFA, provides credit rating analysis of high-yield bonds using external credit ratings as a foundation. At the end of the last quarter, Quanta’s firm, North Investment Bank, held a large position in the bonds of Veyron Corporation, a real estate company with all of its land holdings in a country recently downgraded by several credit rating agencies. The downgrades made Veyron bonds extremely difficult to sell because the bond price has dropped every day since the downgrades. Quanta has been asked by his supervisor to contact the firm’s institutional clients to convince them Veyron bonds are still

an attractive purchase, especially at these lower prices. Quanta does not consider the Veyron bonds a buy at this price level. According to the CFA Institute Code of Ethics and Standards of Professional Conduct, the *most* appropriate action for Quanta is to:

- A obey his supervisor's request.
- B ignore his supervisor's request.
- C promote the bonds with appropriate disclosures.

B is correct because Quanta must refuse to promote Veyron bonds until they are an attractive purchase based on fundamental analysis and market pricing. If Quanta followed the request from his supervisor, he would be in violation of Standard I(B)–Independence and Objectivity, as he does not rate Veyron bonds as a buy. His opinion of the Veyron bonds must not be affected by internal pressure or compensation.

A is incorrect because Quanta should refuse to follow his supervisor's request and promote the bonds as his opinion of the Veyron bonds must not be affected by internal pressure or compensation.

C is incorrect because Quanta should refuse to promote the bonds as his opinion of the Veyron bonds must not be affected by internal pressure or compensation.

Guidance for Standards I–VII

LOS b

Standard I(B)–Independence and Objectivity

- 18 A large manufacturing company is seeking help finding a fund manager for its pension plan. After a comprehensive but unsuccessful search, Brett Arun, CFA, is hired to solicit proposals from various fund managers. The client pays Arun a lump sum fee for his services. The search concludes with Rampart Investments being hired as the pension plan's manager. A year after Rampart is hired, the pension administrator sends Arun a letter telling him how satisfied the pension trustees are with the services provided by the fund manager. Subsequently, without the plan sponsor's knowledge, Arun receives a payment from Rampart for successfully introducing it to the pension plan under an agreement Arun entered into with Rampart when the initial contact with the fund manager was made. With regard to the payment received, did Arun *most likely* violate the CFA Institute Code of Ethics and Standards of Professional Conduct?

- A No.
- B Yes, because he did not disclose the referral fee to the client.
- C Yes, because he should have refused payment from the fund manager.

C is correct because Arun has violated Standard VI(C)–Referral Fees because he did not disclose the referral fee arrangement with Rampart to his client prior to Rampart being appointed as the client's fund manager. This disclosure is necessary for the client to be able to determine Arun's level of independence and objectivity in recommending Rampart to the fund. If Arun had made proper disclosure, he would be able to accept the payment without violating any Standards.

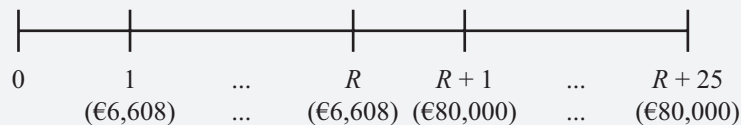
A is incorrect because Arun has violated Standard VI(C)–Referral Fees because he did not disclose the fee paid by Rampart.

B is incorrect because Arun violated Standard VI(C)—Referral Fees when he accepted the fee paid by Rampart. Disclosure after the fact would not cure the acceptance of the fee because disclosure should have occurred prior to the search being conducted. Withholding this information raises the question of a potential lack of objectivity in the recommendations Arun is making. There are also questions concerning the legality of having firms pay so that they may be entered in the money manager search.

Guidance for Standards I–VII
LOS c
Standard VI(C)—Referral Fees

- 19 An individual wants to be able to spend €80,000 per year for an anticipated 25 years in retirement. To fund this retirement account, he will make annual deposits of €6,608 at the end of each of his working years. He can earn 6% compounded annually on all investments. The minimum number of deposits that are needed to reach his retirement goal is *closest* to:
- A 51.
B 40.
C 28.

B is correct. The following figure represents the timeline for the problem:



Using a financial calculator, the funds needed at retirement (R on the timeline) are calculated: $N = 25$; $I/Y = 6\%$; $PMT = €80,000$; Future value (FV) = €0; Mode = End. The calculated present value (PV) is €1,022,668.

$$\begin{aligned}
 PV &= A \left[\frac{1 - \frac{1}{(1+r)^N}}{r} \right] \\
 &= 80,000 \left[\frac{1 - \frac{1}{(1.06)^{25}}}{0.06} \right] \\
 &= 1,022,688
 \end{aligned}$$

Then, €1,022,668 is used as the FV (at R on the timeline) for the accumulation phase annuity as per: $I/Y = 6\%$; $PV = €0$; $PMT = -€6,608$; $FV = €1,022,668$; Mode = End. The computed N is **40**.

Alternatively, 40 could be calculated with the formula:

$$\begin{aligned}
 FV &= A \left[\frac{(1+r)^N - 1}{r} \right] \text{ and solving for } N, \\
 1,022,668 &= 6,608 \left[\frac{(1 + 0.06)^N - 1}{0.06} \right]
 \end{aligned}$$

A is incorrect. 80,000 is multiplied by 25 years (2,000,000) and the result is used as the FV of the 6,608 annuity at 6% and PV = 0. The result is $N = 50.67$.

C is incorrect. 80,000 is multiplied by 25 years and then discounted at 6% for 25 years (465,997). The result is used as the PV of the 6,608 annuity at 6% as follows: $I/Y = 6\%$; $PMT = 6,608$; $PV = 465,997$; $FV = 0$; Calculate N : $N = -28.40$ and the minus sign is ignored.

The Time Value of Money

LOS e, f

Sections 4.1, 6.1

- 20** A company has an unsecured line of credit and needs to maintain its EBIT-to-interest coverage ratio greater than 2.0. Its EBIT is estimated to be between \$36 million and \$48 million, with all values equally likely. If the forecasted interest charge for the year is \$20 million, the probability that EBIT/interest will be more than 2.0 is *closest* to:

- A** 61.5%.
- B** 33.3%.
- C** 66.7%.

C is correct. The EBIT-to-interest ratio is equal to 2.0 when the EBIT is \$40 million. Given that the values between \$36 million and \$48 million are equally likely, the probability of the ratio being equal to or less than 2.0 is 33.3% ($= [\$40 \text{ million} - \$36 \text{ million}] / [\$48 \text{ million} - \$36 \text{ million}]$). Consequently, the probability of the ratio being greater than 2.0 is 66.7% (i.e., $1 - \text{Probability of the ratio being equal to or less than 2.0}$).

A is incorrect. This treats the distribution as discrete with increments in \$1M.

	EBIT	Int	EBIT/INT	
	36	20	1.8	
	37	20	1.85	
	38	20	1.9	
	39	20	1.95	
	40	20	2	
	41	20	2.05	
	42	20	2.1	
	43	20	2.15	
	44	20	2.2	
	45	20	2.25	
	46	20	2.3	
	47	20	2.35	
	48	20	2.4	
Cell Count	13		8	0.615 Prob >2.0

B is incorrect. This is the probability of the ratio being equal to or less than 2.0.

Common Probability Distributions

LOS h

Section 3.1

- 21 A small-cap growth fund's monthly returns for the past 36 months have been consistently outperforming its benchmark. An analyst is determining whether the standard deviation of monthly returns is greater than 6%. Which of the following *best* describes the hypothesis to be tested?
- A $H_0: \sigma^2 \leq 0.36\%$
 B $H_a: \sigma^2 > 6\%$
 C $H_0: \sigma^2 \geq 0.36\%$

A is correct. This is a one-tailed hypothesis testing with a "greater than" alternative hypothesis. A squared standard deviation is being used to obtain a test of variance. The hypotheses are $H_0: \sigma^2 \leq 0.36\%$ versus $H_a: \sigma^2 > 0.36\%$.

B is incorrect as explained in choice A.

C is incorrect as explained in choice A.

Hypothesis Testing
 LOS b
 Section 2

- 22 The variance of returns of Asset A is 625. The variance of returns of Asset B is 1,225. The covariance of returns between Asset A and Asset B is 600. The correlation of returns between Asset A and Asset B is *closest* to:
- A 0.29.
 B 0.69.
 C 0.47.

B is correct. Correlation of returns between asset i and j , $\rho(R_i, R_j)$, is defined as:

$$\rho(R_i, R_j) = \text{Cov}(R_i, R_j) / \sigma(R_i)\sigma(R_j)$$

where

R_i and R_j = the returns of assets i and j

$\text{Cov}(R_i, R_j)$ = the covariance of returns between assets i and j

$\sigma(R_i)$ and $\sigma(R_j)$ = the standard deviations of returns of assets i and j

In this problem, the correlation is $600 / (\sqrt{625} \times \sqrt{1,225}) = 0.6857 \sim 0.69$.

A is incorrect. The mistake is an incorrect order of operations as in $(600 / \sqrt{625}) \times \sqrt{1,225} = 840$, then the square root of 840 is taken: $\sqrt{840} = 28.98$. The result is divided by 100: $28.98/100 = 0.2898 \sim 0.29$.

C is incorrect. It is calculated as follows: $600^2 / (625 \times 1,225) = 0.47$.

Probability Concepts
 LOS k
 Section 3

- 23 The central limit theorem is *best* described as stating that the sampling distribution of the sample mean will be approximately normal for large-size samples:
- A if the population distribution is normal.

- B** for populations described by any probability distribution.
- C** if the population distribution is symmetrical.

B is correct. The central limit theorem holds without regard for the distribution of the underlying population.

A is incorrect because the central limit theorem holds without regard for the distribution of the underlying population.

C is incorrect because the central limit theorem holds without regard for the distribution of the underlying population.

Sampling and Estimation
Section 3.1
LOS e

- 24** Technical analysts *most likely* study trends and patterns in security prices to forecast a company's:
- A** future price trends.
 - B** earnings potential.
 - C** intrinsic value.

A is correct. Technical analysts believe that market trends and patterns tend to repeat, so they rely on recognizing past patterns in an attempt to project future security price patterns.

B is incorrect. Forecasting a company's earnings potential would fall under fundamental analysis, not technical analysis.

C is incorrect. Forecasting a company's intrinsic value would fall under fundamental analysis, not technical analysis.

Technical Analysis
LOS a
Sections 2.1, 2.2

- 25** Given a large random sample, which of the following types of data are *least* appropriately analyzed with nonparametric tests?
- A** Signed data (e.g., number of positives and negatives)
 - B** Ranked data (e.g., 1st, 3rd)
 - C** Numerical values (e.g., 28.43, 79.11)

C is correct. Nonparametric tests are primarily concerned with ranks, signs, or groups, and they are used when numerical parameters are not known or do not meet assumptions about distributions. Even if the underlying distribution is unknown, parametric tests can be used on numerical data if the sample is large.

A is incorrect because nonparametric tests can be used on grouped or counted data.

B is incorrect because nonparametric tests can be used on ranked data.

Hypothesis Testing
LOS k
Section 5

- 26 A group of fund analysts have to select the first, second, and third best fund manager of the year for 2012 based on their subjective judgment. If 10 fund managers are candidates for the three awards, the number of ways in which each analyst can make his ranking is *closest* to:
- A 30.
B 720.
C 120.

B is correct. This problem is a counting one in which order does matter. For this reason, use the permutation formula

$${}_nP_r = \frac{n!}{(n-r)!},$$

where

n = the total number of fund managers; in the problem, $n = 10$.

r = the number of fund managers that will receive the awards (first, second, and third); in the problem, $r = 3$.

$${}_{10}P_3 = \frac{10!}{(10-3)!} = \frac{10!}{7!} = \frac{3,628,800}{5,040} = 720$$

There are 720 ways that each analyst can rank 3 fund managers out of 10, when order does matter.

A is incorrect. It uses the following calculation: $3 \times 10 = 30$.

C is incorrect. It uses the combination formula as follows:

$${}_nC_r = \frac{n!}{(n-r)!r!} = \frac{10!}{(10-3)!3!} = \frac{10!}{7!3!} = \frac{3,628,800}{30,240} = 120$$

Probability Concepts
LOS o
Section 4.2

- 27 The bond-equivalent yield for a semi-annual pay bond is *most likely*:
- A equal to the effective annual yield.
B equal to double the semi-annual yield to maturity.
C more than the effective annual yield.

B is correct. The bond equivalent yield for a semi-annual pay bond is equal to double the semi-annual yield to maturity and is lower than the effective annual yield.

A is incorrect. The bond equivalent yield for a semi-annual pay bond is equal to double the semi-annual yield to maturity and is lower than the effective annual yield.

C is incorrect. The bond equivalent yield for a semi-annual pay bond is equal to double the semi-annual yield to maturity and is lower than the effective annual yield.

Discounted Cash Flow Applications
LOS e, f
Section 4

- 28 An analyst gathers the following information about the performance of a portfolio (\$ millions):

Quarter	Value at Beginning of Quarter (Prior to Inflow or Outflow)	Cash Inflow (Outflow) at Beginning of Quarter	Value at End of Quarter
1	2.0	0.2	2.4
2	2.4	0.4	2.6
3	2.6	(0.2)	3.2
4	3.2	1.0	4.1

The portfolio's annual time-weighted rate of return is *closest* to:

- A 8%.
B 32%.
C 27%.

B is correct. The time-weighted rate of return is calculated by computing the quarterly holding period returns and linking those returns into an annual return as follows:

Quarter	Value (\$ millions) at Beginning of Quarter (Considering Inflows and Outflows)	Value (\$ millions) at End of Quarter	Holding Period Return
1	2.0 + 0.2 = 2.2	2.4	$(2.4 - 2.2)/2.2 = 9.09\%$
2	2.4 + 0.4 = 2.8	2.6	$(2.6 - 2.8)/2.8 = -7.14\%$
3	2.6 - 0.2 = 2.4	3.2	$(3.2 - 2.4)/2.4 = 33.33\%$
4	3.2 + 1.0 = 4.2	4.1	$(4.1 - 4.2)/4.2 = -2.38\%$

The time-weighted return (TWR) is found as follows:

$$\text{TWR} = (1 + 9.09\%) \times (1 - 7.14\%) \times (1 + 33.33\%) \times (1 - 2.38\%) - 1 = 32\% \text{ (rounded)}$$

A is incorrect. It uses the following formula:

$$\frac{[(1 + 9.09\%) \times (1 - 7.14\%) \times (1 + 33.33\%) \times (1 - 2.38\%) - 1]}{4} = 8\% \text{ (rounded)}$$

C is incorrect. It is the money-weighted rate of return (MWR). Solve for r in the following formula:

$$(2 + 0.2) + \frac{0.4}{(1 + r)^1} + \frac{1}{(1 + r)^3} = \frac{0.2}{(1 + r)^2} + \frac{4.1}{(1 + r)^4}$$

To solve, use the cash flow (CF) function of a financial calculator and enter: $CF_0 = -2 - 0.2 = -2.2$, $CF_1 = -0.4$, $CF_2 = 0.2$, $CF_3 = -1$, and $CF_4 = 4.1$. Compute the quarterly IRR and obtain 6.09%. By annualizing we obtain: $(1 + 0.0609)^4 - 1 = 27\%$ (rounded).

Discounted Cash Flow Applications
LOS c, d
Sections 3, 3.2

- 29 The joint probability of events A and B is 32%, with the probability of event A being 60% and the probability of event B being 50%. On the basis of this information, the conditional probability of event A given that event B occurs is *closest to*:
- A 53.3%.
 - B 30.0%.
 - C 64.0%.

C is correct. The conditional probability of A given that B has occurred is equal to the joint probability of A and B divided by the probability of B . In this case:

$$P(A | B) = P(AB)/P(B) = 32.0\%/50.0\% = 64.0\%$$

B is incorrect because it equals $P(A) \times P(B) = 60.0\% \times 50.0\% = 30.0\%$

A is incorrect because it is $P(B | A) = P(AB)/P(A) = 32.0\%/60.0\% = 53.3\%$

Probability Concepts
LOS f
Section 2

- 30 If the stated annual interest rate is 20% and the frequency of compounding is monthly, the effective annual rate (EAR) is *closest to*:
- A 20%.
 - B 21%.
 - C 22%.

C is correct. $EAR = (1 + \text{periodic interest rate})^m - 1 = (1 + 0.20/12)^{12} - 1 = 0.21939\%$, rounded to 22%.

A is incorrect. It is simply the given stated annual rate.

B is incorrect. It uses semiannual compounding as follows: $(1 + 0.20/2)^2 - 1 = 21\%$.

The Time Value of Money
LOS c
Section 3.3

- 31 The following 10 observations are a sample drawn from a normal population: 25, 20, 18, -5, 35, 21, -11, 8, 20, and 9. The mean of the sample is *closest to*:
- A 17.20.
 - B 14.00.

C 15.56.

B is correct. The sum of the 10 numbers is 140. Dividing by 10 gives the mean of 14.

A is incorrect and is calculated by adding the absolute values of the ten numbers (i.e., -11 is valued as 11 and -5 is valued as 5).

C is incorrect and is calculated by dividing 140 by 9 (i.e., by $n - 1$ rather than n).

Statistical Concepts and Market Returns
LOS e
Section 5.1.2

32 A subset of a population is *best* described as a:

- A statistic.
- B sample.
- C conditional distribution.

B is correct. A sample is a subset of a population.

A is incorrect. A statistic is a quantity computed from or used to describe a sample of data.

C is incorrect. A conditional distribution is a misnomer, but it sounds like the distribution of some value conditional upon some event.

Statistical Concepts and Market Returns
LOS a
Section 2.2

33 If the price of a stock goes from \$15.00 to \$16.20 in one year, the continuously compounded rate of return is *closest* to:

- A 7.70%.
- B 8.33%.
- C 8.00%.

A is correct. The continuously compounded rate of return is calculated with the following formula:

$$r_{0,T} = \ln(S_T/S_0)$$

where

$r_{0,T} = r_{0,1}$ = and is the continuously compounded rate of return from time 0 to time T (1 year)

$S_0 = 15.00$ and is the price of the stock at time 0

$S_T = 16.20$ and is the price of the stock at time T (1 year)

The continuously compounded rate of return is: $r_{0,1} = \ln(16.20/15.00) = 7.70\%$.
Alternatively, the end of period price, 16.20 can be found from $15.00 \times e^{0.077 \times 1}$.

B is incorrect. It is calculated as: $\exp(\text{Holding period return}) - 1 = \exp[(16.20/15.00) - 1] - 1 = \exp(8.00\%) - 1 = 8.33\%$.

C is incorrect. It is the holding period return: $(16.20/15.00) - 1 = 8.00\%$.

Common Probability Distributions
LOS o
Section 3.4

- 34 The price of a good falls from \$15 to \$13. Given this decline in price, the quantity demanded of the good rises from 100 units to 120 units. The own-price elasticity of demand for the good is *closest* to:

- A -0.67.
B -1.50.
C -1.25.

B is correct. The own-price elasticity of demand ($E_{P_x}^d$) is calculated as:

$$E_{P_x}^d = \frac{\% \Delta Q_x^d}{\% \Delta P_x} = \frac{\frac{120 - 100}{100}}{\frac{13 - 15}{15}} = -1.50$$

where

$\% \Delta Q_x^d$ = the change in quantity (in %)

$\% \Delta P_x$ = the change in price (in %)

A is incorrect. It uses $\% \Delta P / \% \Delta Q$, calculated as $[(13 - 15)/15] / [(120 - 100)/100] = -0.67$.

C is incorrect. It uses correct formula, but bases $\% \Delta Q$ on new level of demand, calculated as $[(120 - 100)/120] / [(13 - 15)/15] = -1.25$.

Topics in Demand and Supply Analysis
LOS a
Section 2.2

- 35 Three firms operate under perfect competition, producing 900 units of the same product but using different production technologies. Each company's cost structure is indicated in the table:

Company	X	Y	Z
Total Variable Costs	\$2,700	\$3,600	\$4,500
Total Fixed Costs	2,700	1,800	900
Total Costs	\$5,400	\$5,400	\$5,400

Which of the following statements is *most* accurate? If the unit selling price is:

- A \$6.00, all firms should exit the market in the long run.
B \$4.50, all firms should continue to operate in the short run, but exit the market in the long run if these conditions are expected to persist.
C 3.00, Firm X should continue to operate in the short run, but Firms Y and Z should shut down production.

C is correct.

Revenue–Cost Relationship	Short-Run Decision	Long-Term Decision
$TR \geq TC$	Stay in market	Stay in market
$TR > TVC$ but $TR < TFC + TVC$	Stay in market	Exit market
$TR < TVC$	Shut down production	Exit market

TR = Total Revenue; TC = Total Costs; TVC = Total Variable Costs; TFC = Total Fixed Costs

Hence, if the selling price is \$3.00, total revenue for all firms will be $\$3.00/\text{unit} \times 900 \text{ units} = \$2,700$. Only Firm X's variable costs are covered and it should continue operating, while Firms Y and Z should immediately shut down production.

A is incorrect. If total revenue equals or exceeds total costs, the firms should remain in the market both in the long and short run: each firm is just earning an economic profit, which includes its opportunity cost.

B is incorrect. At \$4.50, total revenue is $\$4.50 \times 900 = \$4,050$. Only Firms X and Y cover their variable costs; Firm Z should shut down. It is true, however, that if conditions persist, all should shut down in the long run as they won't be covering their total costs.

Topics in Demand and Supply Analysis

LOS e

Section 3.2.7

36 In a country with a high level of income, as domestic income rises, it is *most likely* that an increase will occur in:

- A** the fiscal balance.
- B** private saving and investment.
- C** the trade balance.

B is correct. In a country with a high level of income, as domestic income rises, private saving and investment will increase.

A is incorrect. The fiscal balance is given by $G - T$. An increase in domestic income leads to an increase in net taxes. Government's fiscal balance will decrease (smaller deficit or larger surplus).

C is incorrect. The trade balance is given by $X - M$. An increase in domestic income leads to an increase in imports and lower net exports. The trade balance will decrease.

Aggregate Output, Prices, and Economic Growth

LOS e

Section 3.1.1

37 In order to reduce a trade deficit, the government of a country experiencing full employment moves to depreciate its currency. As a result, if the country's domestic spending declines relative to income, the *most likely* mechanism that causes this to occur is the:

- A** income effect.

- B** wealth effect.
- C** substitution effect.

B is correct. At full employment, a weaker currency reduces the purchasing power of all domestic currency denominated assets (including the present value of current and future income). Households respond by reducing general expenditures and increasing savings. This response is the wealth effect and reflects the proportion of one's income that is saved (or spent).

A is incorrect. The income effect arises when the price of a good changes: with currency depreciation, foreign goods are more expensive, so real purchasing power (income) is reduced.

C is incorrect. The substitution effect refers to the changes in the composition of spending across different product areas. With currency depreciation, less foreign goods relative to domestic goods are purchased.

Currency Exchange Rates

LOS j

Sections 5.2

Aggregate Output, Prices, and Economic Growth

LOS h

Section 3.3.1

- 38** Which characteristic is a firm *least likely* to exhibit when it operates in a market with a downward sloping demand curve, many competitors, and zero economic profits in the long run?
- A** No pricing power
 - B** Low barriers to entry
 - C** Differentiated product

A is correct. The characteristics of monopolistic competition include a large number of competitors, low pricing power, and the production of differentiated products (through advertising and other non-price strategies), but these still result in some pricing power. The ease of entry results in zero economic profits in the long run.

B is incorrect. Low barriers to entry are a characteristic of monopolistic competition.

C is incorrect. Product differentiation is a characteristic of monopolistic competition.

The Firm and Market Structures

LOS a, b

Sections 2.1, 2.2, 4

- 39** Cost-push inflation is *least likely* to be affected by an increase in:
- A** employee wages.
 - B** finished goods prices.
 - C** commodity prices.

B is correct. Cost-push inflation arises due to increases in costs associated with production: wages and raw materials prices.

A is incorrect. Cost-push inflation arises due to increases in costs associated with production. Wages are the single biggest cost to businesses.

C is incorrect. Cost-push inflation arises due to increases in costs associated with production. Commodities are an input to production, so they are major source of cost-push inflation.

Understanding Business Cycles
LOS h
Section 4.2.4.1

40 The following information is available for 2011:

	New Zealand		Canada	
	Jan 1	Dec 31	Jan 1	Dec 31
Price index	1,137	1,158	117.8	119.9
Nominal exchange rate: NZD/CAD	1.2844			1.2589

The change in the real exchange rate (in NZD/CAD terms) is *closest* to:

- A -2.05%.
- B -1.92%.
- C +1.96%.

A is correct.

	Formula	New Zealand	Canada
Inflation rate	$(I_1 - I_0)/I_0$	$(1,158 - 1,137)/1,137 = 1.85\%$	$(119.9 - 117.8)/117.8 = 1.78\%$
Nominal Exchange Rate Change	$(S_1 - S_0)/S_0$	$(1.2589 - 1.2844)/1.2844 = -1.99\%$	
Real Exchange Rate Change	$\frac{\left(1 + \frac{\Delta S_{\text{NZD/CAD}}}{S_{\text{NZD/CAD}}}\right) \times \left(1 + \frac{\Delta P_{\text{Canada}}}{P_{\text{Canada}}}\right)}{\left(1 + \frac{\Delta P_{\text{NZ}}}{P_{\text{NZ}}}\right)} - 1$	$\frac{(1 - 0.0199) \times (1 + 0.0178)}{(1 + 0.0185)} - 1 = -2.05$	

Approximation:

$$-1.99\% + 1.78\% - 1.85\% = -2.06\%$$

B is incorrect. It uses the correct change in nominal exchange rate, but inverts the inflation rate ratio: $(1 - 0.0199) \times (1.0185)/(1.0178) - 1 = -1.92\%$ or $-1.99\% + 1.85\% - 1.78\%$.

C is incorrect. It uses the appreciation of the Canadian dollar but the correct ratio of inflation rate changes; CAD/NZD: $(0.7943 - 0.7786)/0.7786 = +2.02\%$ or $[1/(1 - 0.0199)] - 1 = 2.03\%$. Giving $(1 + 0.0203) \times 1.0178/1.0185 - 1 = +1.96\%$ or $(+2.03 + 1.78\% - 1.85\% = +1.96\%)$.

Understanding Business Cycles
LOS g
Sections 4.2.2 and 4.2.3
Currency Exchange Rates

(continued)

LOS a, c
Section 2

- 41 Assume that the central bank reduces the reserve requirement. The *most likely* effect will be:
- A a decrease in the money supply.
 - B a decrease in new deposits.
 - C an increase in the money multiplier.

C is correct. Reducing the reserve requirement will increase the money supply, money multiplier, and new deposits.

A is incorrect because the money supply will increase.

B is incorrect because new deposits will increase.

Monetary and Fiscal Policy
LOS c
Section 2.1.2

- 42 Assuming its trading partner does not retaliate, which of the following conditions *must* hold in order for a large country to increase its national welfare by imposing a tariff?
- A The deadweight loss must be smaller than the benefit of its improving terms of trade.
 - B It must auction the import licenses for a fee to offset the decline in the consumer surplus.
 - C It must have a comparative advantage in the production of the imported good.

A is correct. The large country is able to cause the foreign exporter to reduce price in order to retain market share. In the large country, domestic producers gain from higher volume and the government gains from collecting the tariff. The sum of these two gains must exceed the deadweight loss to domestic consumers to achieve a national welfare gain. The change in terms of trade causes income redistribution from the foreign exporter to the domestic producer.

B is incorrect. An import license relates to a quota, not a tariff.

C is incorrect. If the large country had a comparative advantage, it would be exporting more than importing. This is not relevant to whether there is a net domestic gain from the tariff. The tariff hurts domestic consumers. Unless the gain from the tariff exceeds the loss to consumers, national welfare will decrease.

International Trade and Capital Flows
LOS e
Section 3.1

- 43 A dealer report includes the following exchange rate details:

	Spot Rate	Expected Change over Next Year
USD/EUR	1.30	1.75%
CAD/USD	0.95	-0.25%
CHF/EUR	1.22	0.75%

The expected CAD/CHF cross rate in one year is *closest to*:

- A 1.04.
- B 0.98.
- C 1.02.

C is correct.

	Spot Rate	Expected Appreciation	Expected Spot Rate in One Year
USD/EUR	1.30	1.75%	1.323
CAD/USD	0.95	-0.25%	0.948
CHF/EUR	1.22	0.75%	1.229
CAD/CHF	=	$(\text{USD/EUR}) \times [(\text{CAD/USD}) / (\text{CHF/EUR})]$	= 1.020

A is incorrect because the cross rates are calculated incorrectly:

$$\begin{aligned}
 \text{CAD/CHF} &= S_{\text{CHF}}^{\text{CAD}} \times \left[1 + \sum (\text{expected appreciation of the 3 exchange rates}) \right] \\
 &= 1.012 \times [1 + (1.75\% - 0.25\% + 0.75\%)] \\
 &= 1.012 \times (1 + 2.25\%) \\
 &= 1.035
 \end{aligned}$$

B is incorrect. It calculates the Swiss Franc (CHF) relative to the Canadian dollar (CAD): $(\text{CHF/EUR}) / [(\text{USD/EUR}) / (\text{CAD/USD})] = 1.229 / (1.323 / 0.948) = 0.881$.

Currency Exchange Rates

LOS d

Section 3.2

44 Higher than expected inflation will *most likely* lead to an increase in:

- A the real wealth of borrowers.
- B investment.
- C the information content of market prices for economic agents.

A is correct. Unexpected inflation that is higher than anticipated will likely result in borrowers benefiting at the expense of creditors as the real value of their borrowing declines.

B is incorrect because higher than expected inflation will likely result in greater inflation uncertainty. Lenders will ask for a premium to compensate for this uncertainty, which leads to higher borrowing costs and thereby will discourage investment.

C is incorrect because one of the potentially destabilizing effects of unexpected inflation is that it can reduce the information content of market prices.

Monetary and Fiscal Policy
LOS g
Section 2.3.1

45 The structural deficit is equal to the budget deficit:

- A adjusted for inflation.
- B that would exist at full employment.
- C excluding the impact of automatic stabilizers.

B is correct. The structural deficit is the deficit that would exist if the economy was at full employment (or full potential output). Economists often consider the structural deficit as an indicator of the fiscal policy stance.

A is incorrect because the structural deficit makes no adjustment for inflation.

C is incorrect because the structural deficit includes (rather than excludes) the impact of automatic stabilizers on the budget assuming full employment.

Monetary and Fiscal Policy
LOS s
Section 3.3.1

46 All else being equal, if the purchase price of inventory is increasing, a company that accounts for its inventory under last-in, first-out (LIFO) instead of first-in, first-out (FIFO) is *most likely* to have a:

- A higher debt-to-equity ratio.
- B lower net cash flow from operating activities.
- C lower market valuation of its common equity.

A is correct. With rising costs of inventory, a company using LIFO compared with FIFO will report a higher cost of sales and lower profits. This scenario will result in lower increments to retained earnings and a higher debt-to-equity ratio.

B is incorrect. A company using LIFO will report lower taxes paid and a higher net cash flow from operating activities.

C is incorrect. The higher cash flows under LIFO due to lower income taxes paid will increase its market value relative to an identical company that uses FIFO.

Inventories
LOS l, d
Sections 3.7, 4

47 Which of the following *best* describes common equity?

- A The initial investment by common shareholders in the company
- B The resources owned or controlled by a company
- C The residual interest in a company's assets after deducting its liabilities

C is correct. Common equity is a component of the balance sheet and represents the owners' residual interest in the company's assets after deducting its liabilities.

A is incorrect. Common equity includes the initial investment by the shareholders and the retained earnings; this definition is incomplete.

B is incorrect. Assets are a component of the balance sheet and represent resources controlled by an enterprise as a result of past events and from which future economic benefits to the enterprise are expected to flow.

Understanding Balance Sheets
LOS a
Section 2

- 48 The cumulative amount of earnings recognized on a company's income statements that have not been distributed as dividends to the company's owners is *best* described as:

- A retained earnings.
- B accumulated other comprehensive income.
- C dividends payable.

A is correct. Retained earnings, a component of equity, is defined as the cumulative amount of earnings recognized on the company's income statements that have not been distributed as dividends to the company's owners.

B is incorrect. Other comprehensive income is also a component of equity, but it is defined as items of comprehensive income not reported on the income statement.

C is incorrect. Dividends payable represent dividends that have been declared but not yet paid. Not all earnings not yet paid out would be accrued as a liability; only those that have been declared as dividends.

Understanding Balance Sheets
LOS f
Section 6.1

- 49 Income statements for two companies (A and B) and the common-size income statement for the industry are provided in the following table:

(\$ thousands)	Company A	Company B	Industry
Sales	\$10,500	\$8,250	100.0%
Cost of goods sold	6,353	5,239	62.8%
Selling, general, and administrative expenses	2,625	2,021	24.8%
Interest expense	840	536	7.0%
Pretax earnings	683	454	5.4%
Taxes	205	145	1.7%
Net earnings	\$478	\$309	3.7%

The *best* conclusion an analyst can make is that:

- A Company A earns a higher gross margin than both Company B and the industry.

- B** both companies' tax rates are higher than the industry average.
C Company B's interest rate is lower than the industry average.

A is correct. Common-sized analysis of the income statements shows that Company A has a lower percentage cost of goods sold and thus a higher gross margin than the industry and Company B.

	Company A	Company B	Industry	Company A	Company B
Sales	\$10,500	\$8,250	100.0%	100%	100%
Cost of goods sold	6,353	5,239	62.8%	60.5%	63.5%
Gross margin			37.2%	39.5%	36.5%

Company A earns a higher gross margin than both Company B and the industry.

	Company A	Company B	Industry	Company A	Company B
Pretax earnings	\$683	\$454	5.4%	6.5%	5.5%
Taxes	205	145	1.7%	2.0%	1.8%
Tax rate = Taxes/ Pretax earnings			32%	30%	32%

The tax rates for the companies are not higher than the industry.

The tax rates for the companies are not higher than the industry. The interest rate is not a function of sales and cannot be analyzed on a common-size income statement. Tax rates are determined based on Taxes/Pretax earnings, not as a percentage of sales (as shown in common-size analysis).

B is incorrect. Tax rate varies with the pretax income, and as shown in the table neither companies' tax rate is above the industry average.

C is incorrect. Although Company B's interest rate as a percentage of sales is lower than the industry average, interest cost does not vary with sales, but with the level of debt and its risk, so the actual interest rate cannot be determined.

Understanding Income Statements

LOS k

Section 7

Financial Analysis Techniques

LOS b, c

Sections 3.1, 3.2.2

- 50** The following selected balance sheet and ratio data are available for a company:

Metric	Current Year	Previous Year
Cash and cash equivalents	98.0	
Marketable securities	389.2	
Accounts receivables	12.0	
Other current assets	120.1	
Total current assets	619.3	
Deferred revenues	85.0	
Other current liabilities	92.3	

Metric	Current Year	Previous Year
Total current liabilities	177.3	
Cash ratio		2.37
Quick ratio		2.97
Current ratio		3.27

Which of the following ratios *most likely* decreased this year?

- A Quick
- B Current
- C Cash

A is correct.

Metric	Current Year	Previous Year	Conclusion
Cash ratio = (Cash + Marketable securities)/Current liabilities	$(98 + 389.2)/177.3 = 2.75$	2.37	Increase
Quick ratio = (Cash + Marketable securities + Receivables)/Current liabilities	$(98 + 389.2 + 12)/177.3 = 2.82$	2.97	Decrease
Current ratio = Current assets/Current liabilities	$619.3/177.3 = 3.49$	3.27	Increase

B is incorrect. The current ratio increased in the current year.

C is incorrect. The cash ratio increased in the current year.

Understanding Balance Sheets
LOS h
Section 7.2
Financial Analysis Techniques
LOS b
Section 4.3

51 The following items are from a company's cash flow statement.

Classification of Cash Flow	Description	Amount (£ thousands)
Operating activities	Cash received from customers	55,000
Investing activities	Interest and dividends received	10,000
Financing activities	Net repayment of revolving credit loan	12,000

Which of the following standards and formats did the company *most likely* use in the preparation of its financial statements?

- A Either IFRS or US GAAP, direct format
- B IFRS, indirect format
- C IFRS, direct format

C is correct. The direct method of cash flow statement presentation shows the specific cash inflows and outflows that result in reported cash flow from operating activities (e.g., cash from customers and cash to suppliers). Companies using IFRS can decide to report interest and dividend receipts as either an investing or operating activity; under US GAAP, they must report such income as an operating activity. The listed operating and investment activities indicate that the company reports under IFRS using the direct method.

A is incorrect. The description of investing activities, which includes interest and dividends received, indicates that the company must be reporting under IFRS. Those items are operating activities under US GAAP.

B is incorrect. The description of operating activities indicates that the company is using the direct method.

Understanding Cash Flow Statements
LOS c, d
Sections 2.3, 2.3.2, 3.2.1.5

- 52 If a company capitalizes an expenditure related to capital assets instead of expensing it, ignoring taxes, the company will *most likely* report:
- A a lower cash flow per share in that period.
 - B the same free cash flow to the firm (FCFF) in that period.
 - C a higher earnings per share in future periods.

B is correct. The FCFF [Cash flow from operations (CFO) + Interest $\times$ (1 - t) - Capital expenditures] would be the same. CFO and capital expenditures would both increase by the same amount (ignoring taxes). Therefore, the net effect on FCFF would be zero.

Example Capitalizing delivery cost as opposed to expensing it

Ignoring taxes		
FCFF	CFO + Interest $\times$ (1 - t) - Capital expenditures	
	Capital expenditures	If capitalized, the amount capitalized increases capital expenditures and is recorded as a cash outflow from investing activities
	CFO	The CFO will be higher by amount capitalized (i.e., the amount not expensed)

Because capital expenditures and CFO increase by the same amount, ignoring taxes, FCFF is unchanged.

A is incorrect. As indicated, CFO increases in the period of capitalization. Cash flow per share is based on CFO, so it would increase.

C is incorrect. Future EPS will be lower due to the higher future depreciation expense arising from the current capitalization.

Understanding Cash Flow Statements
LOS i
Section 4.3
Long-Lived Assets
LOS a
Section 2.1

- 53 A company has recently revalued one of its depreciable properties and estimates that its remaining useful life will be another 20 years. The applicable tax rate for all years is 30%, and the revaluation of the property is not recognized for tax purposes. Details related to this asset are provided in the following table:

Original Values and Estimates (millions)	Accounting Purposes	Tax Purposes
Acquisition cost in 2011	£8,000	£8,000
Depreciation, straight line	20 years	8 years
Accumulated depreciation, end of 2013	£1,200	£3,000
Net balance, end of 2013	£6,800	£5,000
Re-estimated Values and Estimates, Start of 2014		
Revaluation balance, start of 2014	£10,000	Not applicable
New estimated life	20 years	

The deferred tax liability related to this asset (in millions) as at the end of 2014 is *closest* to:

- A £960.
 B £690.
 C £1,650.

B is correct.

(millions)	Accounting Purposes	Tax Purposes
Revaluation surplus	(£10,000 – £6,800) = £3,200	No revaluation allowed
Depreciation, straight line	20 years	5 years remaining
Start of year balance after revaluation, 2013	£10,000	£5,000
Depreciation, 2013	(£10,000/20 years) = £500	£1,000
Net balance, end of 2013	£9,500	£4,000
Minus revaluation surplus	– £ 3,200	—
Carrying value for purposes of deferred taxes	£6,300	£4,000

$$\text{Deferred tax liability} = 0.30 \times (£6,300 - £4,000) = £690$$

Only the portion of the difference between the tax base and the carrying amount that is not the result of the revaluation is recognized as giving rise to a deferred tax liability. The portion arising from the revaluation surplus is used to reduce the revaluation surplus in equity.

A is incorrect. It is the tax rate $\times$ writeup = $0.30 \times 3,200 = 960$.

C is incorrect. It is the tax rate $\times$ the difference between carrying value and tax base: $0.30 \times (9,500 - 4,000) = 1,650$.

- 54 The year-end balances in a company's last-in, first-out (LIFO) reserve are \$56.8 million as reported in the company's financial statements for both 2013 and 2014. For 2014, the measure that will most likely be the same regardless of whether the company uses the LIFO or the first-in, first-out (FIFO) inventory method is the:
- A gross profit margin.
 - B amount of working capital.
 - C inventory turnover.

A is correct. The LIFO reserve did not change from 2013 to 2014. With no change in the LIFO reserve, cost of goods sold will be the same under both methods. Sales are always the same for both methods, so gross profit margin will be the same for 2014. The FIFO inventory will be higher because the LIFO inventory and LIFO reserve are added to compute FIFO inventory. Because the inventory balances would differ under FIFO, both inventory turnover and the amount of working capital would also differ under FIFO.

B is incorrect. The FIFO inventory would be higher because the LIFO inventory and LIFO reserve are added to compute FIFO inventory. Because the inventory balances would be different under FIFO, net working capital would be different under FIFO.

C is incorrect. The FIFO inventory would be higher because the LIFO inventory and LIFO reserve are added to compute FIFO inventory. Because the inventory balances would be different under FIFO, the inventory turnover would be different under FIFO.

Inventories
LOS e, k, l
Section 4.1

- 55 The following selected fixed asset information is available for a company:

	2016 (\$US millions)
Cost: Total property, plant, and equipment (PP&E)	30,815
Accumulated depreciation	16,465
Net PP&E	14,350
Average net PP&E	12,200
Net sales	21,670
Net income	2,705

The company's fixed asset turnover ratio is *closest* to:

- A 1.78.
- B 8.01.
- C 1.51.

A is correct. The fixed asset turnover ratio for the company is calculated as
 $\text{Net sales} / \text{Average net PP\&E} = 21,670 / 12,200 = 1.78$

B is incorrect. It mistakenly uses net income instead of the net sales: $21,670 / 2,705 = 8.01$.

C is incorrect. It mistakenly uses net PP&E year-end instead of the average:
 $21,670/14,350 = 1.51$.

Long-Lived Assets
 LOS m
 Section 7

- 56 Which of the following ratios is *most likely* to be used as a measure of operating performance?
- A Cash ratio
 - B Working capital turnover ratio
 - C Defensive interval ratio

B is correct. Activity ratios are typically used to measure operating performance. Working capital turnover is an example of an activity ratio; the defensive interval ratio and cash ratio are liquidity ratios used to measure a company's ability to meet its short-term obligations.

A is incorrect. The cash ratio is an example of a liquidity ratio.

C is incorrect. The defensive interval ratio is an example of a liquidity ratio.

Financial Analysis Techniques
 LOS b
 Sections 4.2.1, 4.3.1

- 57 The following information is available about a company (\$ millions):

Year Ended 31 December	2012	2011
Sales	322.8	320.1
Net income	27.2	26.8
Cash flow from operations	15.3	38.1

During 2012, the company *most likely* experienced a significant decrease in:

- A inventory, anticipating lower demand for its products in 2013.
- B the proportion of sales made on a cash basis.
- C the proportion of interest-bearing debt relative to trade accounts payable.

B is correct. Sales are nearly the same for the two years. A decrease in the proportion of cash sales implies an increase in the proportion of credit sales, which would increase accounts receivable and decrease cash flow from operations.

A is incorrect. A decrease in inventory would increase cash from operations.

C is incorrect. An increase in payables would increase cash from operations.

Understanding Cash Flow Statements
 LOS h
 Sections 2.3.1, 3.2.5

- 58 Selected information from a company's recent income statement and balance sheets is presented in the following table.

Selected Financial Information as of 31 December

(C\$ thousands)	2013	2012
Sales	2,240,000	
Cost of goods sold (COGS)	1,320,000	
Cash and investments	210,700	191,600
Accounts receivable	212,800	201,900
Inventories	63,000	71,500
Accounts payable	129,600	157,200
Other current liabilities	130,700	182,700

The company operates in an industry in which suppliers offer terms of 2/10, net 30. The payables turnover for the average company in the industry is 8.5 times. Which of the following statements is *most* accurate? In 2013, the company, on average:

- A paid its accounts within the payment terms provided.
- B paid its accounts more promptly than the average firm in the industry.
- C took advantage of early payment discounts.

B is correct. The firm's days in payables is 39.9 days (see following calculations), so it appears that the firm does not normally take supplier provided discounts (paying in 10 days) or pay its accounts within the 30-day terms provided. However, on average, the company is paying faster than the average firm in the industry (42.9 days).

Payables turnover = Purchases/Average payables = 1,311,500/143,400 = 9.15 times

where:

$$\begin{aligned}\text{Purchases} &= \text{COGS} + \text{End inventory} - \text{Beginning inventory} \\ &= 1,320,000 + (63,000 - 71,500) \\ &= 1,311,500\end{aligned}$$

$$\begin{aligned}\text{Average payables} &= (129,600 + 157,200)/2 \\ &= 143,400\end{aligned}$$

Days in payables = 365/Payables turnover ratio

For firm: 365 days/9.15 = 39.9 days

For industry: 365 days/8.5 times = 42.9 days

A is incorrect. On average, the company is exceeding the maximum terms provided of 30 days.

C is incorrect. The firm on average does not take advantage of the early discounts offered or its days in payables would be less than 10 days.

Financial Analysis Techniques

LOS b

Sections 4.2.1, 4.2.2

Working Capital Management

LOS f

Section 2.2

59 The following information is available on a company:

Metric	
Fixed charge coverage ratio required by debt covenant	3.50
Forecasted interest expense (\$ thousands)	800
Forecasted lease payments (\$ thousands)	300
Tax rate	30%

The minimum net income (in thousands) that the company must generate to meet its debt covenant requirement is *closest* to:

- A \$2,750.
- B \$1,925.
- C \$2,135.

B is correct. The fixed charge coverage ratio is:

$$\frac{\text{Net income} + \text{Income tax expense} + \text{Interest expense} + \text{Lease payments}}{\text{Interest payments} + \text{Lease payments}}$$

	Calculation	\$ thousands
Denominator (\$ thousands): Interest payments + Lease payments	800 + 300	1,100
Minimum numerator: Net income + Income tax expense + Interest expense + Lease payments	Denominator × 3.50	3,850
Minus interest expense		-800
Minus lease payments		-300
Income before tax		2,750
Minus income tax expense	2,750 × 30%	-825
Minimum net income		1,925

A is incorrect. This answer incorrectly omits tax expense from the numerator. The answer provided is actually net income before tax as calculated above, not net income.

C is incorrect. This answer incorrectly assumes that the numerator in the ratio is EBIT.

(\$ thousands)

Minimum numerator (from above): Incorrectly assumed to be EBIT	Denominator × 3.50	3,850
Less: interest expense		-800
Incorrect income before tax		3,050
Less: income tax expense	\$3,050 × 30%	-915
Incorrect minimum net income		2,135

Financial Analysis Techniques
LOS b
Section 4.4.1
Non-Current (Long-term) Liabilities

(continued)

LOS k
Section 5

- 60 Which of the following *best* describes a reason a company would acquire the use of equipment through an operating lease rather than by purchase?
- A To take advantage of less costly financing
 - B To obtain preferential tax treatment for the lease payments compared with ownership
 - C To increase cash from operations

A is correct. Leases can provide less costly financing. Because of the tax and economic advantages enjoyed by lessors, they are often able and willing to offer attractive lease terms resulting in less costly financing to the lessees.

B is incorrect. Lessors (the owners) are normally in a better position to take advantage of tax deductions, such as depreciation and interest.

C is incorrect. Cash from operations would be lower with an operating lease compared to purchasing the asset.

Non-Current (Long-Term) Liabilities
LOS f
Section 3.1

- 61 An analyst examining the statement of cash flows for possible manipulation is *least likely* to be concerned about a(n):
- A cash flow from operations to net income ratio consistently higher than 1.
 - B increase in cash from operations arising from a large change in accounts payable.
 - C change in the classification of interest paid from an operating cash flow to a financing cash flow.

A is correct. A cash flow from operations to net income ratio that is consistently higher than 1 indicates that operating cash flow is consistently higher than net income and signals high earnings quality.

B is incorrect. A large increase in accounts payable could mean that a company is trying to artificially increase cash flow from operations by delaying payments to creditors.

C is incorrect. Although this change is allowable, it increases cash flow from operations, making the company appear healthier, and is likely to be made for this reason.

Financial Reporting Quality
LOS i
Section 4.2.2

- 62 A company that prepares its financial statements in accordance with IFRS issues £5,000,000 face value 10-year bonds on 1 January 2013 when market interest rates for such bonds are 5.50%. The bonds carry a coupon of 6.50% with interest paid annually on 31 December. The carrying value of the bonds as of 31 December 2014 will be *closest* to:

- A £5,316,000.
 B £4,695,000.
 C £5,301,000.

A is correct. There are two ways to determine the value of the bonds on 31 December 2014.

First method:

Calculate the present value (PV) of the cash flows over the remaining eight years at 5.5%:

$$£5,000,000 \times 6.5\% \times PVA(8 \text{ years}, 5.5\%) + £5,000,000 \times PV(8 \text{ years}, 5.5\%) = £5,316,728$$

Or using a financial calculator:

PMT = £325,000, $i = 5.5\%$, $n = 8$ years, Future value = £5,000,000. Compute PV; PV = £5,316,728.

Second method:

Determine the initial bond proceeds and then the amortization of the premium or discount during the first two years. The initial bond proceeds are determined using a financial calculator:

PMT = £325,000, $i = 5.5\%$, $n = 10$ years, Future value = £5,000,000. Compute PV; PV = £5,376,881.

Using the effective annual interest rate method, which is required under IFRS, to amortize the premium gives the following:

Year	Carrying Amount at Start of Year	Interest Expense at EAI	Interest Payment at Coupon Rate	Amortization of Premium	Carrying Amount at End of Year
2013	5,376,881	295,728	325,000	29,272	5,347,609
2014	5,347,609	294,119	325,000	30,881	£5,316,728

B is incorrect. It confuses the interest expense as the coupon payment and the payment based on the EAI in the valuation of the bond, so that bond value is at a discount: i.e., 4,640,558 and the discount is amortized

Year	Carrying Amt at Start of Year	Interest Expense at EAI	Interest Payment at Coupon Rate	Amortization of Discount	Carrying Amount at End of Year
2013	4,640,558	301,636	275,000	29,272	4,667,194
2014	4,667,194	303,368	275,000	30,881	4,695,562

C is incorrect. It uses the straight-line method to amortize the premium.

$$\begin{aligned} \text{Annual Amortization of Premium} &= (5,376,881 - 5,000,000)/10 \text{ years} \\ &= 376,881/10 \\ &= 37,688 \end{aligned}$$

After 2 years: Carrying Value = 5,376,881 - 2 × 37,688 = 5,301,505.

Non-Current (Long-Term) Liabilities

LOS a, b

Sections 2.1, 2.2

- 63 Under the International Accounting Standards Board's (IASB's) Conceptual Framework, one of the qualitative characteristics of useful financial information is that different knowledgeable users would agree that the information is a faithful representation of the economic events that it is intended to represent. This characteristic is *best* described as:
- A understandability.
 - B verifiability.
 - C comparability.

B is correct. Under the IASB's Conceptual Framework, verifiability means that different knowledgeable and independent users would agree that the information presented faithfully represents the economic events that it is intended to represent.

A is incorrect. Understandability is the clear and concise presentation of information.

C is incorrect. Comparability allows users to identify and understand similarities and differences of items.

Financial Reporting Standards
LOS d
Section 5.2

- 64 The following information about a company is provided:

Account	\$ thousands
Contributed capital, beginning of the year	50
Retained earnings, beginning of the year	225
Sales revenues earned during the year	450
Investment income earned during the year	5
Total expenses paid during the year	402
Dividends paid during the year	10
Total assets, end of the year	800

Total liabilities (in \$ thousands) at the end of the year are *closest* to:

- A 482.
- B 487.
- C 472.

A is correct. Given $\text{Assets} = \text{Liabilities} + \text{Equity}$, first calculate ending equity (\$318, see calculation in the following table).

$$\$800 = \text{Liabilities} + \$318$$

$$\text{Total liabilities} = \$482$$

	\$ thousands
Contributed capital	50
Initial retained earnings	225
Sales revenues	450

	\$ thousands	
Investment income	5	
Total expenses	(402)	
Net income for the year	53	
Dividends paid	(10)	
Increase in retained earnings	43	43
Ending owners' equity		\$318

B is incorrect. It forgets to include the investment income.

$$\begin{aligned}\text{Ending equity} &= 450,000 - 402,000 - 10,000 + 50,000 + 225,000 = 313,000 \\ 800,000 - 313,000 &= 487,000\end{aligned}$$

C is incorrect. It fails to deduct the dividends.

$$\begin{aligned}\text{Ending equity} &= 318,000 + 10,000 = 328,000 \\ \text{Total liabilities} &= 800,000 - 328,000 = 472,000\end{aligned}$$

Financial Reporting Mechanics
LOS c, f
Sections 3.2, 4.2

- 65 Common-size income statements are shown for three companies in the same industry. Which company is *most likely* to have a technically superior product?

	Company X	Company Y	Company Z
Revenue	100%	100%	100%
Cost of goods sold	65	50	30
Administrative expenses	20	20	20
Research and development	0	5	30
Advertising expenses	5	15	10
Operating profit	10	10	10

- A Company Z
B Company Y
C Company X

A is correct. Company Z has spent the most on research and development and is able to support the highest gross margin (lowest cost of goods sold). It likely has the technically superior product.

B is incorrect. Company Y spends significantly less than Company Z on research and development. It is unlikely to have a technically superior product.

C is incorrect. Company X spends nothing on research and development. It is unlikely to have a technically superior product.

Understanding Income Statements
LOS k
Section 7.1

- 66 Management's commentary (also known as management's discussion and analysis) *most likely* includes:
- A supplementary information about accounting policies, methods, and estimates.
 - B an auditor's opinion as to the fair presentation of the financial statements.
 - C a discussion of significant trends, events, and uncertainties that affect the operating results.

C is correct. Management's commentary includes a discussion of significant trends, events, and uncertainties that affect the operating results.

A is incorrect. The notes disclose information about the accounting policies, methods, and estimates used to prepare the financial statements.

B is incorrect. The Auditor's Report includes the auditor's opinion as to the fair presentation of the financial statements.

Financial Statement Analysis: An Introduction
LOS c
Section 3.1.6

- 67 A US company that complies with US GAAP would like to exclude some items in determining non-GAAP financial measures, other than EBIT and EBITDA. Which of the following items may be excluded?
- A For performance measures, items tagged as infrequent that occurred within the past two years
 - B Impairment charges for long-lived assets
 - C For liquidity measures, litigation costs requiring cash settlement

B is correct. To assist investors in evaluating operating performance, companies often report non-GAAP earnings by excluding asset impairment charges either for long-lived assets, goodwill, or other intangible assets.

A is incorrect because the SEC prohibits the calculation of a non-GAAP performance measure intended to eliminate or smooth items tagged as non-recurring. The period within two years either before or after the reporting date is the relevant time frame for considering whether a charge or gain is a recurring item.

C is incorrect because the SEC prohibits the exclusion of charges or liabilities requiring cash settlement from any non-GAAP liquidity measures other than EBIT and EBITDA.

Financial Reporting Quality
LOS g
Section 4.1

- 68 Which of the following conditions is *most likely* associated with decreased earnings quality? Compared with the prior year, the reporting entity's earnings:
- A decreased slightly in response to the introduction of conservative accounting policies.
 - B were similar in magnitude but included a large gain on the sale of a manufacturing plant.

- C increased slightly because of a reduction in bad debt expense based on more-current experiences.

B is correct. The sale of a manufacturing plant is likely a one-time transaction that will not be sustained in future years. The quality of reported earnings has therefore decreased from the prior year.

A is incorrect. This is an example of decreased financial reporting quality because conservatism, a choice made by management, is making it more difficult to establish expectations for the future. Since the earnings only decreased because of the conservatism, there is no decrease in the underlying earnings quality.

C is incorrect. If the estimates are based on more recent experiences, it does not imply the intent to manipulate earnings and will provide a more faithful representation of the company's performance.

Financial Reporting Quality
LOS a
Sections 1 and 2.2

- 69 An analyst is comparing the financial leverage of two companies, A and B, from the same industry.
- Both companies can borrow at a rate of 4%.
 - The two companies are virtually identical except that Company A leases essentially all of its premises; Company B owns all of its premises.
 - Company A recorded €15,280 (thousand) of lease expenses in 2015, the current year, ending 31 December. The following excerpt is from the notes to its 2015 financial statements:

Note on Leasing Activities: Non-Cancellable Operating Lease Rentals Are Payable on 1 January as Follows:

	€ thousands
2016	15,280
2017	15,280
2018	15,280

To facilitate a fair comparison with Company B, the analyst will *most likely* adjust (in € thousands) for the operating leases by increasing Company A's:

- A earnings before tax by €15,280.
B liabilities by €45,840.
C liabilities by €44,100.

C is correct. Analysts typically adjust for operating leases by treating them as if they were finance leases, including them as a liability measured at present value of future lease payments. In this case, the future lease payments are an annuity due of €15,280 over three years, at 4%. The present value of the annuity is €44,100: $15,280 \times PVA^{ADV}(\text{three years, 4\%}) = €44,100$.

A is incorrect. This is the nominal value of one lease payment. The analyst would adjust net income by the difference between the operating lease expense and the estimated amount of interest expense and depreciation expense that would be deducted under financing lease treatment.

B is incorrect. This is the nominal value of the future lease payments: $3 \times €15,280 = €45,840$.

Financial Statement Analysis: Applications
LOS e
Section 6.6

70 The following information is available for a firm:

Market risk premium	7.0%
Risk-free rate	2.0%
Comparable firm return	10.4%
Comparable firm debt-to-equity ratio	1.0
Comparable firm tax rate	40.0%

The firm's unleveraged beta is *closest* to:

- A 0.75.
- B 1.20.
- C 1.05.

A is correct. Find the comparable firm's beta: $(10.4\% - 2.0\%) / 7.0\% = 1.20$.

Unlever the comparable firm's beta: $\beta_{L, \text{comparable}} / [1 + (1 - \text{Tax rate}) \times \text{Debt-to-equity ratio}]$

$$1.20 / [1 + (1 - 40\%) \times 1.0] = 0.75.$$

B is incorrect because it is the comparable firm's beta.

C is incorrect because the comparable firm's beta is computed as $1.68 = (10.4\% - 2.0\%) / (7.0\% - 2.0\%)$ and then proceeds to the next calculation.

Cost of Capital
LOS h, i
Section 3.3.1, 4.1

71 The acceptance of which of the following capital budgeting projects is *most likely* to expose a company to the highest level of uncertainty?

- A Replacement of worn out equipment
- B Expansion projects
- C Newly launched product or services



C is correct. Investments related to new products or services expose the company to even more uncertainties than expansion projects. These decisions are more complex and will involve more people in the decision-making process.

A is incorrect. Replacement of worn out equipment is simply an improvement to the existing project with recurring revenues.

B is incorrect. Investments related to new products or services expose the company to even more uncertainties than expansion projects. These decisions are more complex and will involve more people in the decision-making process

Capital Budgeting

LOS a

Section 2

72 The optimal capital budget for a firm is *best* described as occurring when the company's marginal cost of capital is:

- A** equal to the investment opportunity schedule.
- B** less than the investment opportunity schedule.
- C** greater than the investment opportunity schedule.



A is correct. The optimal capital budget occurs when the marginal cost of capital (MCC) intersects with (is equal to) the investment opportunity schedule (IOS).

B is incorrect. The optimal capital budget occurs when the marginal cost of capital (MCC) intersects with (is equal to) the investment opportunity schedule (IOS).

C is incorrect. The optimal capital budget occurs when the marginal cost of capital (MCC) intersects with (is equal to) the investment opportunity schedule (IOS).

Cost of Capital

LOS d

Section 2.3

73 Financial risk is *least likely* affected by:

- A** debentures.
- B** dividends.
- C** long-term leases.



B is correct. By taking on fixed obligations, such as debt (including debentures) and long-term leases, a company increases its financial risk. Dividends will not increase financial risk.

A is incorrect because the use of debentures (one type of bond) is directly associated with financial risk.

C is incorrect because the use of long-term leases is directly related to financial risk.

Measures of Leverage

LOS a

Section 3.4

- 74 Which of the following is the *least* appropriate method for an external analyst to use to estimate a company's target capital structure for determining the weighted average cost of capital (WACC)?
- A Using the company's current capital structure at book value weights
 - B Using averages of comparable companies' capital structure
 - C Using statements made by the company's management regarding capital structure policy

A is correct. An external analyst does not know a company's actual target capital structure. Consequently, the analyst should rely on market value (not book value) weights for the components of the company's current capital structure.

B is incorrect: This is an accepted method for an external analyst to estimate a company's target capital structure.

C is incorrect: This is an accepted method for an external analyst to estimate a company's target capital structure.

Cost of Capital
LOS c
Section 2.2

- 75 Which is *most likely* considered a secondary source of liquidity?
- A Centralized cash management system
 - B Trade credit
 - C Liquidating long-term assets

C is correct. Liquidating long-term assets is a secondary source of liquidity.

A is incorrect. Centralized cash management system is considered as a primary source of liquidity.

B is incorrect. Trade credit (part of short-term funds) is considered as a primary source of liquidity.

Working Capital Management
LOS a
Sections 2.1.1, 2.1.2

- 76 Under the stakeholder theory, corporate governance is *most* consistent with a system of:
- A internal controls and procedures by which individual companies are managed.
 - B defined roles for management and the majority shareowner(s).
 - C checks and balances to minimize the conflicting interests among shareowners.

A is correct. Corporate governance is the system of internal controls and procedures by which individual companies are managed.

B is incorrect. The majority shareholder doesn't necessarily have a specific role that is defined through corporate governance. Instead, the majority shareholder exercises influence and/or control through voting mechanisms tied to their shareholdings.

C is incorrect. Corporate governance is primarily aimed at managing the conflicting interests between management and external shareholders, not amongst shareholders.

Corporate Governance and ESG: An Introduction
LOS d
Section 2

- 77 Two mutually exclusive projects have the following cash flows (€) and internal rates of return (IRR):

Project	IRR	Year 0	Year 1	Year 2	Year 3	Year 4
A	27.97%	-2,450	345	849	635	3,645
B	28.37%	-2,450	345	849	1,051	3,175

Assuming a discount rate of 8% annually for both projects, the *best* decision for the firm to make is to accept:

- A both projects.
- B Project B only.
- C Project A only.

C is correct. The NPV of project A is €1,780.59.

$$1,780.59 = -2,450 + \frac{345}{(1.08)^1} + \frac{849}{(1.08)^2} + \frac{635}{(1.08)^3} + \frac{3,645}{(1.08)^4}$$

The NPV of Project B is €1,765.36.

$$1,765.36 = -2,450 + \frac{345}{(1.08)^1} + \frac{849}{(1.08)^2} + \frac{1,051}{(1.08)^3} + \frac{3,175}{(1.08)^4}$$

Because Project A has a higher NPV and the projects are mutually exclusive, only Project A should be accepted.

A is incorrect because both projects cannot be accepted when the projects are mutually exclusive.

B is incorrect because Project A has a higher NPV.

Capital Budgeting
LOS c, d
Sections 4.1, 4.2, 4.8

- 78 An investment strategy that focuses on climate change is *most likely* following which approach to environmental, social, and governance (ESG) investing?

- A Thematic
- B Best in class
- C Impact



A is correct. A strategy that considers a single factor, such as climate change, is a thematic investment strategy.

B is incorrect. Best-in-class ESG investing focuses on identifying the best ESG scoring companies in each industry.

C is incorrect. Impact investing attempts to achieve targeted social or environmental objectives.

Corporate Governance and ESG: An Introduction

LOS k, j

Section 9.3

79 An investment policy statement's risk objective states that over a 12-month period, with a probability of 95%, the client's portfolio must not lose more than 5% of its value. This statement is *most likely* a(n):

- A** total risk objective.
- B** relative risk objective.
- C** absolute risk objective.



C is correct. The statement is an absolute risk objective because it expresses a maximum loss in value with an associated probability of loss.

A is incorrect because this is an absolute (not total) risk objective because it expresses a maximum loss in value with an associated probability of loss.

B is incorrect because this is an absolute (not relative) risk objective because it expresses a maximum loss in value with an associated probability of loss.

Basics of Portfolio Planning and Construction

LOS c

Section 2.2

80 The top level of a risk management system *most likely* is:

- A** risk governance.
- B** strategic analysis or integration.
- C** defined policies or procedures.



A is correct. Normally a role of the board of directors of a company, risk governance is where goals and responsibilities are defined and top-level decisions (such as determining the company's risk tolerance) are made.

B is incorrect because strategic analysis or integration occurs at the end of the process and leads to changes in investment decision-making and firm strategy going forward.

C is incorrect because policies and procedures extend risk governance into the day to day operations of the company.

Risk Management: An Introduction

LOS b

Section 2

- 81 An investor's transactions in a mutual fund and the fund's returns over a four-year period are provided in the following table:

	Year			
	1	2	3	4
New investment at the beginning of the year (US\$)	2,500	1,500	1,000	0
Investment return for the year	-20%	65%	-25%	10%
Withdrawal by investor at the end of the year (US\$)	0	-500	-500	0

Based on this data, the money-weighted return (or internal rate of return) for the investor is *closest* to:

- A 2.15%.
 B 7.50%.
 C 3.96%.

C is correct.

Year	1	2	3	4
Starting balance (US\$)	0.00	2,000.00	5,275.00	4,206.25
New investment at the beginning of the year (US\$)	2,500.00	1,500.00	1,000.00	0.00
Net balance at the beginning of year (US\$)	2,500.00	3,500.00	6,275.00	4,206.25
Investment return for the year	-20%	65%	-25%	10%
Investment gain (loss) (US\$)	-500.00	2,275.00	-1,568.75	420.63
Withdrawal by investor at the end of the year (US\$)	0.00	-500.00	-500.00	0.00
Balance at the end of year (US\$)	2,000.00	5,275.00	4,206.25	4,626.88

The money-weighted return is calculated by solving for i in the following equation:

$$2,500 = \frac{-1,500}{(1+i)^1} + \frac{-500}{(1+i)^2} + \frac{500}{(1+i)^3} + \frac{4,626.88}{(1+i)^4}$$

$$CF_0 = -2,500$$

$$CF_1 = -1,500 \text{ (new investment beginning of Year 2)}$$

$$CF_2 = -500 \text{ (withdrawal of 500, end of Year 2; -1,000 new investment beginning Year 3)}$$

$$CF_3 = -500 \text{ (withdrawal of 500, end of Year 3)}$$

$$CF_4 = 4,626.88 \text{ (balance at end of Year 4)}$$

$$i = 0.0396$$

A is incorrect. This is the geometric mean.

$$[(1 - 0.2) \times (1 + 0.65) \times (1 - 0.25) \times (1 + 0.1)]^{1/4} - 1 = 0.0215$$

B is incorrect. This is the arithmetic mean: $(-0.2 + 0.65 - 0.25 + 0.10)/4 = 0.075$.

Portfolio Risk and Return: Part I
LOS a
Section 2.1.4

- 82 Risk that can be attributed to factor(s) that affect a company or industry is *best* described as:
- A non-systematic risk.
 - B market risk.
 - C systematic risk.

A is correct. Risk that is attributable to company-specific or industry-specific factors is referred to as non-systematic risk.

B is incorrect. Market risk is the same as systematic risk. It is due to factors that impact the whole market.

C is incorrect. Systematic risk is the same as market risk. It is due to factors that impact the whole market.

Portfolio Risk and Return: Part II
LOS c
Section 3.1

- 83 As one moves to the right along an investor's efficient frontier, a set increase in risk is most likely to lead to:
- A sequentially smaller increases in expected return.
 - B consistent increases in expected return.
 - C sequentially larger increases in expected return.

A is correct. The increase in return with every unit increase in risk keeps decreasing as one moves from left to right because the slope of the efficient frontier continues to decrease. Thus, investors obtain decreasing increases in returns as they assume more risk.

B is incorrect. The slope of the efficient frontier continues to decrease, leading to smaller incremental returns, not consistent.

C is incorrect. The slope of the efficient frontier continues to decrease, leading to smaller incremental returns, not larger.

Portfolio Risk and Return: Part I
LOS g
Section 5.2

- 84 Information about a portfolio that consists of two assets is provided below:

Asset	Portfolio Weight	Standard Deviation
A	25%	12%
B	75%	16%

If the correlation coefficient between the two assets is 0.75, the standard deviation of the portfolio is *closest* to:

- A 15.00%.
- B 12.37%.
- C 14.39%.

C is correct.

$$[(0.25^2 \times 0.12^2) + (0.75^2 \times 0.16^2) + (2 \times 0.25 \times 0.75 \times 0.12 \times 0.16 \times 0.75)]^{0.5} = 0.1493 = 14.39\%$$

A is incorrect. It omits the correlation coefficient in calculating the standard deviation: $[(0.25^2 \times 0.12^2) + (0.75^2 \times 0.16^2) + (2 \times 0.25 \times 0.75 \times 0.12 \times 0.16)]^{0.5} = 0.15 = 15\%$, which is the weighted average standard deviation.

B is incorrect. It omits the third term in the formula in calculating the standard deviation: $[(0.25^2 \times 0.12^2) + (0.75^2 \times 0.16^2)]^{0.5} = 0.1237 = 12.37\%$.

Portfolio Risk and Return: Part I
LOS e
Sections 4.1.2, 4.1.3

85 In general, which of the following institutions will *most likely* have a high need for liquidity and a short investment time horizon?

- A Banks
- B Defined-benefit pension plans
- C Endowments

A is correct. Banks have a short-term horizon and high liquidity needs.

B is incorrect. Defined-benefit pension plans have a long time horizon and low need for liquidity.

C is incorrect. Endowments have a long time horizon and low need for liquidity.

Portfolio Management: An Overview
LOS b
Section 3

86 If the expected return on the market portfolio is 6% and the risk-free rate is 2%, the expected return of a security with a beta of 1.25 is *closest* to:

- A 7.00%.
- B 5.00%.
- C 9.50%.

A is correct. The capital asset pricing model posits that the expected return of a security is $E(R_i) = R_f + \beta_i[E(R_m) - R_f]$ where R_f is the risk-free rate, R_m is the return on the market portfolio, and β is the beta of the security: $2\% + 1.25 \times (6\% - 2\%) = 7.00\%$.

B is incorrect because it fails to add the risk-free rate: $1.25 \times (6\% - 2\%) = 5.00\%$.

C is incorrect because it fails to subtract the risk-free rate from the market portfolio's expected return: $2\% + 1.25 \times (6\%) = 9.50\%$.

Portfolio Risk and Return: Part II

LOS g

Section 3.2.6

- 87 If the following three stocks are held in a portfolio, the portfolio's total return on an equal-weighted basis is *closest* to:

Stock	Number of Shares Owned	Beginning of Period Price per Share (\$)	End of Period Price per Share (\$)	Dividend per Share during the Period (\$)
A	500	40	37	2.00
B	320	50	52	1.50
C	800	30	34	0.00

- A 3.28%.
B 5.94%.
C 6.37%.

B is correct. Equal weighting assigns an equal weight to each constituent security at inception. Therefore, it is the sum of the total return from each security divided by the number of securities in the portfolios.

Stock	$(P_1 - P_0 + D)/P_0$	Total Return (%)
A	$(37 - 40 + 2.00)/40 =$	-2.5
B	$(52 - 50 + 1.50)/50 =$	7.00
C	$(34 - 30 + 0)/30 =$	13.33
Portfolio return with equal weighting: $(-2.50 + 7.00 + 13.33)/3 =$		5.94

A is incorrect. It is computed on a price return basis not total return.

Stock	Price Return (%) = $(P_1 - P_0)/P_0$
A	$(37 - 40)/40 = -7.50$
B	$(52 - 50)/50 = 4.00$
C	$(34 - 30)/30 = 13.33$
Portfolio return with equal weighting: $(-7.50 + 4.00 + 13.33)/3 = 3.28\%$	

C is incorrect. It is the total return on the basis of beginning of period market-capitalization weights.

Stock	Total Return (%) $= (P_1 - P_0 + D)/P_0$	BOP Weights*	Total Return (%) $\times$ BOP Weight
A	$(37 - 40 + 2.00)/40 = -2.50$	0.333	-0.83
B	$(52 - 50 + 1.50)/50 = 7.00$	0.267	1.87
C	$(34 - 30 + 0)/30 = 13.33$	0.400	5.33
Portfolio market-capitalization-weighted total return =			6.37

* BOP weights:
Beginning Value of Portfolio: $(A = 500 \times \$40) + (B = 320 \times \$50) + (C = 800 \times \$30) = \$60,000$;
BOP weights: $A = 20,000/60,000 = 0.333$; $B = 16,000/60,000 = 0.267$; $C = 24,000/60,000 = 0.400$

Security Market Indexes
LOS d, e
Section 3.2.2

- 88 If the number of financial analysts who follow or analyze a company increases substantially, then the market for this company's shares will *most likely* become:
- A more attractive for active investors.
 - B overvalued.
 - C more efficient.

C is correct. The number of financial analysts who follow or analyze a security or asset should be positively related to market efficiency. Therefore, if more analysts cover a company, the market for this company's shares will most likely become more efficient.

A is incorrect. In a more efficient market, less profitable trading opportunities exist and as a consequence, it becomes less attractive for active investors.

B is incorrect. In a more efficient market, prices should converge toward fair value.

Market Efficiency
LOS c
Section 2.3

- 89 A corporate manager pursuing a low-cost strategy will *most likely*:
- A engage in offering products of unique quality or type.
 - B have strong market research teams for product development and marketing.
 - C invest in productivity-improving capital equipment.

C is correct. A corporate manager pursuing a cost leadership strategy must be able to invest in productivity-improving capital equipment for achieving cost controls and being able to offer products and services at lower prices than the competition.

A is incorrect. Offering products that are unique either in quality, type, or means of distribution is suitable for differentiation strategies.

B is incorrect. Having strong market research teams for product development and marketing is suitable for differentiation strategies.

Introduction to Industry and Company Analysis
LOS k
Section 6

- 90 For a US investor, which of the following statements concerning investing in depository receipts (DRs) is *least* accurate?
- A Investing in DRs could provide arbitrage opportunities and entail currency risk.
 - B Investors in unsponsored DRs would have the same voting rights as the direct owners of common shares.
 - C Sponsored DRs are subject to greater reporting requirements than unsponsored DRs.



B is correct. Investors of unsponsored DRs would not have the same voting rights as the direct owners of common shares because the depository bank retains the voting rights.

A is incorrect because it is an accurate statement. The DRs trading on multiple exchanges could experience short-term valuation discrepancies, potentially giving rise to a quick arbitrage profit opportunity for astute traders to exploit. The price of each DR will be affected by factors that affect the price of the underlying shares and exchange rate movements.

C is incorrect because it is an accurate statement. Sponsored DRs are subject to greater reporting requirements than unsponsored DRs. In the United States, sponsored DRs must be registered with the SEC.

Overview of Equity Securities
LOS a, b
Section 5.2

- 91 Industry analysis is *least* useful to those who are engaged in:
- A a top-down investment approach.
 - B indexing and passive investing strategies.
 - C portfolio performance attribution.



B is correct. Indexing and passive investing strategies would not engage in over- or underweighting of industries, industry rotation, or timing investments in industries. Therefore, industry analysis is not useful to such investors or portfolio managers.

A is incorrect. In a top-down investing approach, industry analysis is useful to identify industries with positive, neutral, or negative outlooks for profitability and growth, which will then help weighting of industries relative to the benchmark.

C is incorrect. Portfolio performance attribution, which addresses the sources of a portfolio's returns, usually in relation to the portfolio's benchmark, includes industry or sector selection.

Introduction to Industry and Company Analysis
LOS a
Section 2

- 92 Which of the following statements about the forms of market efficiency is *least* accurate? If the form of market efficiency is:
- A weak, then investment strategies based on fundamental analysis could achieve abnormal returns.
 - B semi-strong, then security prices fully reflect all past market data.
 - C strong, then prices reflect only private information.

C is correct. If markets are strong-form efficient, prices reflect not only private information but also past market data and public information. If markets are weak-form efficient, investment strategies based on fundamental analysis of public information and past market data could achieve abnormal returns. The semi-strong-form of market efficiency also encompasses the weak form. Therefore, security prices reflect not only publicly known and available information but also all past market data.

A is incorrect. This statement is correct because if markets are weak-form efficient only, investment strategies based on fundamental analysis could achieve abnormal returns.

B is incorrect. This statement is correct because the semi-strong-form of market efficiency also encompasses the weak-form; therefore, security prices reflect not only publicly known and available information but also all past market data.

Market Efficiency
LOS e, d
Section 3

- 93 An investor borrows the maximum amount allowed by the initial margin requirement of 40% to purchase 100 shares of a stock selling at \$60 per share. If the investor sells the stock when its price increases to \$70 per share, her return before commissions and interest will be *closest* to:
- A 41.7%.
 - B 27.8%.
 - C 16.7%.

A is correct.

$$\begin{aligned}
 \text{Investor's return (\%)} &= \frac{\text{Market value of the stock} - \text{Loan}}{\text{Investor's equity}} - 1 \\
 &= \frac{(70 \times 100) - (60 \times 100 \times 0.6)}{(60 \times 100 \times 0.4)} - 1 \\
 &= 41.67\%
 \end{aligned}$$

B is incorrect It takes margin as 60%.

$$= \frac{(70 \times 100) - (60 \times 100 \times 0.4)}{(60 \times 100 \times 0.6)} - 1$$

$$= 27.78\%$$

C is incorrect It ignores the margin.

$$= \frac{100 \times (70 - 60)}{100 \times 60}$$

$$= 16.67\%$$

Market Organization and Structure
LOS f
Section 5

94 The following market information relates to a company:

Market price per share	\$37.80
Number of shares outstanding	1,000,000
Net income	\$5,250,000
Total common equity	\$35,000,000
Total annual dividend paid	\$1,512,000
Risk-free rate	2.60%
Market risk premium	8.00%
Beta	1.05

Using the capital asset pricing model (CAPM), the company's cost of equity is closest to:

- A 15.0%.
- B 12.4%.
- C 11.0%.

C is correct. Using the CAPM:

$$\begin{aligned} \text{Cost of equity} &= \text{Risk-free rate} + (\text{Beta} \times \text{Market risk premium}) \\ &= 2.6 + (1.05 \times 8) = 11\% \end{aligned}$$

B is incorrect. It is the Dividend yield + (Beta × Market risk premium)

$$\text{Dividend per share} = 1,512,000 / 1,000,000 = 1.512$$

$$\text{Dividend yield} = 1.512 / 37.8 = 4\%$$

$$\text{Cost of equity (with the mistake)} = 4\% + (1.05 \times 8) = 12.4\%$$

A is incorrect. It is the Return on equity = Net income/Common equity

$$\begin{aligned} \text{ROE} &= 5,250,000 / 35,000,000 \\ &= 15\% \end{aligned}$$

Overview of Equity Securities
LOS h
Section 7.2

Equity Valuation: Concepts and Basic Tools
LOS g
Section 4

- 95 Which of the following statements about peer groups is *most* accurate? A peer group is constructed through a process:
- A that starts with an existing commercially classified system that is then narrowed.
 - B that locates a group of companies whose valuation is influenced by diverse factors.
 - C where management should refrain from participating to maintain objectivity in the process.

A is correct. The process consists of initially examining commercial classification systems and then refining it to the companies operating in the chosen industry.

B is incorrect because the process includes looking for similar, not diverse, factors.

C is incorrect because management should be part of the process.

Introduction to Industry and Company Analysis
LOS d
Section 4.4

- 96 Assume the current dividend of a security is \$9.50. The dividend is expected to grow by 12% each year for two years and then 3% afterwards. The required rate of return is 15%. The security's value is *closest* to:
- A \$95.58.
 - B \$120.51.
 - C \$94.99.

A is correct. The value of the security is:

$$D_0 = \$9.50$$

$$D_1 = \$9.50 \times (1 + 0.12) = \$10.64$$

$$D_2 = \$9.50 \times (1 + 0.12)^2 = \$11.92$$

$$D_3 = \$9.50 \times (1 + 0.12)^2 \times (1 + 0.03) = \$12.27$$

$$V_2 = \frac{\$12.27}{0.15 - 0.03}$$

$$V_0 = \frac{\$10.64}{(1 + 0.15)} + \frac{\$11.92}{(1 + 0.15)^2} + \frac{\$102.25}{(1 + 0.15)^2} \cong \$95.58$$

B is incorrect; it takes the discounted values from V_0 but doesn't discount D_3 .

$$V_0 = \frac{\$10.64}{(1 + 0.15)} + \frac{\$11.92}{(1 + 0.15)^2} + \$102.25 \cong \$120.51$$

$$9.25 + 9.01 + \$102.25 = \$120.51$$

C is incorrect; it takes the security value and adds dividend 0 (D_0): $85.49 + \$9.50 = \94.99 .

Equity Valuation: Concepts and Basic Tools
LOS g
Section 4.3

- 97 Which of the following *best* describes an advantage of the EV/EBITDA multiple for valuing equity? An advantage is that:
- A the multiple must be positive.
 - B it does not require the market value of debt.
 - C EBITDA is a proxy for operating cash flow.

C is correct. An advantage of EBITDA is that it is a proxy for operating cashflow because it excludes depreciation and amortization.

A is incorrect. The multiple can be negative if EBITDA is negative.

B is incorrect. The market value of debt is needed in valuing the equity since it should be deducted from the enterprise value and may be difficult to obtain.

Equity Valuation: Concepts and Basic Tools
LOS m
Section 5.4

- 98 Which of the following transactions is *most likely* to affect a company's financial leverage ratio?
- A Payment of a 9% stock dividend
 - B An increase in cash dividends paid
 - C Completion of a previously announced 1-for-20 reverse stock split

B is correct. Cash dividends affect a company's capital structure and financial leverage ratios by reducing assets and shareholders' equity.

A is incorrect. Neither stock splits nor stock dividends affect a firm's financial leverage ratio.

C is incorrect. Neither stock splits nor stock dividends affect a firm's financial leverage ratio.

Equity Valuation: Concepts and Basic Tools
LOS a
Section 4.1

- 99 Credit spreads are *most likely* to narrow during:
- A economic contractions.
 - B a period of flight to quality.
 - C economic expansions.



C is correct. Credit spreads narrow during economic expansions and widen during economic contractions. During an economic expansion, corporate revenues and cash flows rise, making it easier for corporations to service their debt, and investors purchase corporates instead of Treasuries, causing spreads to narrow.

A is incorrect. Credit spreads narrow during economic expansions and widen during economic contractions.

B is incorrect. During a flight to quality investors sell corporate and buy treasuries thereby widening the credit spread on corporates.

Fundamentals of Credit Analysis

LOS h

Section 6

100 Which of the following is *least likely* a short-term funding method available to banks?

- A** Central bank funds
- B** Negotiable certificate of deposits
- C** Syndicated loans



C is correct. A syndicated loan is a loan from a group of lenders, called the "syndicate," to a single borrower. Syndicated loans are primarily originated by banks, and the loans are extended to companies but also to governments and government-related entities.

A is incorrect because central bank funds are one of the short-term wholesale funds available to banks for short-term funding needs.

B is incorrect because a negotiable CD allows any depositor (initial or subsequent) to sell the CD in the open market prior to the maturity date. CDs are an important source of funds for financial institutions.

Fixed-Income Markets: Issuance, Trading, and Funding

LOS h

Section 7

101 In the securitization process, which of the following is *most likely* a third party to the transaction? The:

- A** seller of the collateral.
- B** special purpose entity.
- C** financial guarantor.



C is correct. In the securitization process, the seller of the collateral, the special purpose entity, and the servicer of the loan are the main parties. All other parties, including independent accountants, lawyers/attorneys, trustees, underwriters, rating agencies, and financial guarantors are third parties to the transaction.

A is incorrect because in the securitization process the seller of the collateral is one of the main parties to the transaction.

B is incorrect because in the securitization process the special purpose entity is one of the main parties to the transaction.

Introduction to Asset-Backed Securities
LOS b
Section 3.2

102 Consider bonds that have the same yield to maturity and maturity. The bond with the greatest reinvestment risk is *most likely* the one selling at:

- A** a premium.
- B** par.
- C** a discount.



A is correct. Yield to maturity is based on the assumption that a bond is held to maturity, does not default, and has its coupon payments reinvested at the yield to maturity. The bond selling at a premium has the highest coupon rate and is expected to earn the most reinvestment income from reinvesting those coupon payments at the yield to maturity. If the reinvestment rate falls, this bond will suffer the greatest loss.

B is incorrect because the bond selling at par has a lower coupon rate than the bond selling at a premium.

C is incorrect because the bond selling at a discount has a lower coupon rate than the bond selling at a premium.

Understanding Fixed-Income Risk and Return
LOS a
Section 2

103 On 15 December 2013, Alpha Corp. issued a 10-year callable bond paying an annual coupon of 8%. The bond is callable in whole or in part at any time after 15 December 2018. This type of callable bond is *most likely* referred to as:

- A** American style.
- B** European style.
- C** Bermuda style.



A is correct. An American-style callable bond is a bond in which the issuer has the right to call the bonds at any time starting on the first call date.

B is incorrect because in a European-style callable bond the issuer has the right to call the bonds only once on the call date.

C is incorrect because a callable bond where the issuer has the right to call the bonds on specified dates after the call protection period has elapsed is a Bermuda-style callable bond.

Fixed-Income Securities: Defining Elements
LOS f
Section 3.1

- 104 In a rising interest rate environment, the effective duration of a puttable bond relative to an otherwise identical non-puttable bond, will *most likely* be:
- A higher.
 - B lower.
 - C the same.



B is correct. When interest rates are rising, the put option becomes more valuable to the investor. The ability to sell the bond at par value limits the price depreciation as rates rise. So, the presence of an embedded put option reduces the sensitivity of the bond price to changes in interest rates, resulting in a lower effective duration.

A is incorrect because in a rising interest rate environment the effective duration of a puttable bond will be lower, not higher, than the effective duration of a comparable non-puttable bond.

C is incorrect because in a rising interest rate environment the effective duration of a puttable bond will be lower than the effective duration of a comparable non-puttable bond.

Understanding Fixed-Income Risk and Return
LOS f
Section 3.3

- 105 Which of the following is *least likely* to be a negative covenant associated with a coupon-paying corporate bond issue?
- A A requirement to pay withholding taxes to foreign governments in a timely manner
 - B A prohibition from investing in long-term projects in emerging market countries
 - C A requirement to hedge at least 50% of the firm's revenues generated from foreign sales



A is correct. Requiring compliance with the existing rules and regulations of foreign governments is administrative in nature and thus an affirmative covenant.

B is incorrect because this is a negative covenant that is likely to materially constrain the firm's operational decisions and is likely to be costly to the firm.

C is incorrect because this is a negative covenant that is likely to materially constrain the firm's operational decisions and is likely to be costly to the firm.

Fixed-Income Securities: Defining Elements
LOS c
Section 3.1

- 106 The bonds of Apex Corporations have a par value of \$10,000 each and an annual required rate of return of 10%. The bonds make quarterly coupon payments at an annual rate of 6% and have two years remaining until maturity. The current market price of each bond is *closest* to:
- A \$10,749.
 - B \$9,283.
 - C \$9,306.

B is correct. Using the quarterly coupon payment of \$150 [= (0.06 × 10000)/4] over eight quarters and a quarterly required rate of return of 2.5%, we calculate the bond's price as:

$$P_0 = 150/(1.025)^1 + 150/(1.025)^2 + \dots + 150/(1.025)^8 + 10,000/(1.025)^8 = \$9,282.99$$

A is incorrect because the bond's price is computed using a quarterly coupon payment of \$250 [= (0.10 × 10000)/4] and a quarterly required rate of return of 1.5%:

$$P_0 = 250/(1.015)^1 + 250/(1.015)^2 + \dots + 250/(1.015)^8 + 10000/(1.015)^8 = \$10,748.59$$

C is incorrect because the bond's price is computed using the annual coupon payment of \$600 [= (0.06 × 10,000)] over 2 years and the annual required rate of return of 10%:

$$P_0 = 600/(1.10)^1 + (600 + 10,000)/(1.10)^2 = \$9,305.79$$

Introduction to Fixed-Income Valuation
LOS a
Section 2.1

107 Which type of fixed-income security is *most likely* to have coupon payments that reset periodically?

- A** Callable bonds
- B** Floating-rate notes
- C** Convertible bonds

B is correct. A floating-rate bond does not have a fixed coupon rate over its life. Instead, its coupon payments reset periodically according to some reference rate, such as the one-month London interbank offered rate (Libor).

A is incorrect because a callable bond gives issuers the ability to retire debt prior to maturity. It does not have a feature of resetting coupon payments periodically according to the reference rate.

C is incorrect because a convertible bond gives the bondholder the right to convert the bond into a specified number of shares of the issuer's common shares.

Fixed-Income Markets: Issuance, Trading, and Funding
LOS b, g
Sections 2.1 and 6.3

108 The semiannual bond equivalent yield spot rates for US Treasury yields are provided below.

Period	Years	Spot Rate
1	0.5	1.20%
2	1.0	2.10%
3	1.5	2.80%
4	2.0	3.30%

On a semiannual bond equivalent yield (BEY) basis, the six-month forward rate one year from now is *closest to*:

- A 4.21%.
- B 3.64%.
- C 2.10%.

A is correct. The x -year forward rate y years from now is

$${}_x f_y = \frac{(1 + z_{x+y})^{x+y}}{(1 + z_y)^y} - 1$$

All spot rates are given on a BEY basis and must be divided by 2 in this calculation:

$${}_{0.5} f_{1.0} = \frac{[1 + (0.028/2)]^3}{[1 + (0.021/2)]^2} - 1 = 0.021036$$

On a BEY basis, the forward rate is $0.021036 \times 2 = 4.21\%$.

B is incorrect because it calculates the 1-year forward rate six months from now.

C is incorrect because it doesn't convert to a semiannual bond equivalent yield basis.

Introduction to Fixed-Income Valuation
LOS h
Section 4

- 109 Treasury spot rates on a semiannual bond equivalent yield basis are provided below.

Maturity	Semiannual Bond Equivalent Yield
0.5 years	0.40%
1.0 years	0.80%
1.5 years	1.00%
2.0 years	1.10%
2.5 years	1.20%

Using these spot rates, the value of a 2.5-year Treasury security that makes semiannual payments based on a 2% coupon rate is *closest to*:

- A 101.98.
- B 106.88.
- C 99.06.

A is correct. The value of the bond is

$$\frac{1}{1.002^1} + \frac{1}{1.004^2} + \frac{1}{1.005^3} + \frac{1}{1.0055^4} + \frac{101}{1.006^5} = 0.9980 + 0.9920 + 0.9851 + 0.9783 + 98.0238$$

$$= 101.98$$

B is incorrect because it discounts all cash flows at the 2.5-spot rate and uses the annual coupon rate instead of the semi-annual:

$$\frac{2}{1.006^1} + \frac{2}{1.006^2} + \frac{2}{1.006^3} + \frac{2}{1.006^4} + \frac{102}{1.006^5} = 1.9881 + 1.9762 + 1.9644 + 1.9527 + 98.9943 = 106.88$$

C is incorrect because it fails to divide the spot rates by 2 as required because they are presented on a bond-equivalent yield basis:

$$\frac{1}{1.004^1} + \frac{1}{1.008^2} + \frac{1}{1.01^3} + \frac{1}{1.011^4} + \frac{101}{1.012^5} = 0.9960 + 0.9842 + 0.9706 + 0.9572 + 95.1522 = 99.06$$

Introduction to Fixed-Income Valuation
LOS c
Section 2.4

110 A bond's duration is 7.31, and its convexity is -24.85 . Using the duration model with convexity adjustment, the bond's percentage change in price if interest rates decrease 2% is *closest to*:

- A 15.12%.
- B 15.60%.
- C 14.12%.

C is correct. The duration model estimates the percentage change in price as $-\text{AnnModDur} \times \Delta\text{Yield}$, or $-7.31 \times (-0.02) = +14.62\%$, and the convexity adjustment is $\frac{1}{2} \times \text{AnnConvexity} \times (\Delta\text{Yield})^2$, or $\frac{1}{2} \times (-24.85) \times (0.02)^2 = -0.50\%$, and $14.62\% - 0.50\% = 14.12\%$.

A is incorrect because it uses $-C \times \Delta i$ as the convexity adjustment or $-24.85 \times -0.02 = 0.50\%$ and $14.62\% + 0.50\% = 15.12\%$.

B is incorrect because it uses $-C \times (\Delta i)^2 \times 100$ or $-24.85 \times (0.02)^2 \times 100 = +0.98\%$ as the convexity adjustment and $14.62\% + 0.98\% = 15.60\%$.

Understanding Fixed-Income Risk and Return
LOS i
Section 3.6

111 For a forward contract with a value of zero, a situation where the spot price is above the forward price is *best* explained by high:

- A interest rates.
- B storage costs.
- C convenience yield.

C is correct. If the convenience yield is high, holding the underlying confers large benefits, thus the spot price can exceed the forward price for a forward contract with a value of zero. Based on the formula

$$V_t(T) = S_t - (\gamma - \theta)(1 + r)^t - F_0(T)(1 + r)^{-(T-t)}$$

and an initial value $V_t(0)$ of zero, large benefits γ explain why the spot price can exceed the forward price.

A is incorrect. High interest rates make the forward contract more valuable. Thus the forward rate is above the spot rate.

B is incorrect. High storage costs make the forward contract more valuable. Thus the forward rate is above the spot rate.

Basics of Derivative Pricing and Valuation

LOS d

Section 2.2.5

112 According to put–call–forward parity, if the put in a protective put with forward contract expires out of the money, the payoff is *most likely* equal to:

- A** the market value of the underlying asset.
- B** zero.
- C** the face value of a risk-free bond.



A is correct. A protective put with forward contract is defined as a long position in (1) a bond that has the face value equal to the forward contract, (2) a forward contract, and (3) a long position in a put. If the put expires out of the money, the value of the overall position is equal to the market value of the asset.

$+ F_0(t)$	(payoff of bond)
$+ S_T - F_0(t)$	(payoff of forward)
$+ 0$	(payoff of option)
$= S_T$	(payoff of strategy)

B is incorrect. Zero is the payoff of the put alone. This ignores the other positions in the strategy.

C is incorrect. The face value of the risk-free bond is the payoff of the protective put with forward contract if the put expires in the money

Basics of Derivative Pricing and Valuation

LOS m

Section 4.1.10

113 A derivative can *best* be described as a financial instrument that:

- A** duplicates the underlying asset's performance.
- B** transforms the underlying asset's performance.
- C** passes through the underlying asset's returns.



B is correct. The best characterization of a derivative is that it typically transforms the underlying asset's performance.

A is incorrect. A derivative transforms the performance of the underlying asset rather than duplicating the performance of the underlying asset.

C is incorrect. A derivative transforms the performance of the underlying asset rather than passing through the returns of the underlying asset.

Derivative Markets and Instruments
LOS a
Section 2

114 In a credit default swap, the party that receives a series of cash payments in return for promising to pay compensation for credit losses resulting from a third party's default is *most likely* the:

- A** clearinghouse.
- B** seller of the swap.
- C** buyer of the swap.



B is correct. A credit default swap is a derivatives contract between a credit protection buyer and a credit protection seller in which the seller receives a series of cash payments from the buyer in return for a promise of compensation for credit losses resulting from a third party's default.

A is incorrect. A credit default swap is a derivatives contract between a credit protection buyer and a credit protection seller in which the seller (not the clearinghouse) receives a series of cash payments from the buyer in return for a promise of compensation for credit losses resulting from a third party's default.

C is incorrect. A credit default swap is a derivatives contract between a credit protection buyer and a credit protection seller in which the seller (not the buyer) receives a series of cash payments from the buyer in return for a promise of compensation for credit losses resulting from a third party's default.

Derivative Markets and Instruments
LOS c
Section 4.2.2

115 A swap that involves the exchange of a fixed payment for a floating payment is *most likely* equivalent to a series of:

- A** off-market forward contracts.
- B** forward contracts that all have an initial positive value.
- C** forward contracts that all have an initial value equal to the fixed payment.



A is correct. Because the cost of carrying an asset over different time periods will vary, the values of the implicit forward contracts embedded in the swap will not be equal: some may be positive, and some may be negative. Off-market forward contracts satisfy this condition because they can be set at any value.

B is incorrect. Because the initial market value of the swap is zero by definition, it cannot be replicated by a series of forward contracts with an initial positive value.

C is incorrect. Because the cost of carrying an asset over different time periods will vary, the prices of the implicit forward contracts embedded in the swap cannot all be equal.

Basics of Derivative Pricing and Valuation
LOS g
Section 3.3

116 Which of the following *most likely* belongs in an alternative asset category?

- A** A limited partnership that takes long and short positions in publicly traded equity.
- B** Equity in an emerging market company that is traded over-the-counter.
- C** Securitized commercial real estate debt.

A is correct. A limited partnership that takes long and short positions in publicly traded equity is one type of hedge fund, a category of alternative assets.

B is incorrect because traded equity, even equity that is traded over the counter, is a part of the traditional equity asset category.

C is incorrect because securitized real estate debt (i.e., CMBS and RMBS) are part of the publicly traded debt universe, which is not an alternative asset.

Introduction to Alternative Investments
LOS b
Section 2.1

117 High Plains Capital is a hedge fund with a portfolio valued at \$475,000,000 at the beginning of the year. One year later, the value of assets under management is \$541,500,000. The hedge fund charges a 1.5% management fee based on the end-of-year portfolio value as well as a 10% incentive fee. If the incentive fee and management fee are calculated independently, the effective return for a hedge fund investor is *closest* to:

- A** 12.29%.
- B** 10.89%.
- C** 11.06%.

B is correct.

$$\text{Management fee} = \$541,500,000 \times 0.015 = \$8,122,500$$

$$\text{Incentive fee} = (\$541,500,000 - \$475,000,000) \times 0.10 = \$6,650,000$$

$$\text{Total fees} = \$14,772,500$$

$$\text{Return} = (\$541,500,000 - \$475,000,000 - \$14,772,500) / \$475,000,000 = 0.1089 \text{ or } 10.89\%$$

A is incorrect because only the management fee is included in the return calculation.

$$\text{Return} = (\$541,500,000 - \$475,000,000 - \$8,122,500) / \$475,000,000 = 0.1229 \text{ or } 12.29\%$$

C is incorrect. The incentive fee is incorrect. It is incorrectly calculated as follows:

$$\text{Incentive fee} = (\$541,500,000 - \$475,000,000 - \$8,122,500) \times 0.10 = \$5,837,750$$

$$\text{Total fees} = \$13,960,250 = \$8,122,500 + \$5,837,750$$

$$\text{Return} = (\$541,500,000 - \$475,000,000 - \$13,960,250) / \$475,000,000 = 0.1106 \text{ or } 11.06\%$$

Introduction to Alternative Investments
LOS f
Section 3.3.1

118 Collectibles are *least likely* to provide:

- A** long-term capital appreciation.
- B** portfolio diversification.
- C** current income.

C is correct. Collectibles do not provide current income, but they can potentially provide long-term capital appreciation and help further diversify a portfolio.

A is incorrect. Collectibles can potentially provide long-term capital appreciation.

B is incorrect. Collectibles can potentially provide portfolio diversification.

Introduction to Alternative Investments
LOS d
Section 7

119 Which of the following hedge fund strategies emphasizes a top-down approach?

- A** Macro
- B** Equity hedge
- C** Event-driven

A is correct. Macro hedge funds emphasize a “top down” approach to identify economic trends and trade on expected movements in economic variables.

B is incorrect because equity hedge funds use a “bottom up” approach and employ strategies, such as market neutral, which uses quantitative (technical) and/or fundamental analysis to identify under- and overvalued equity securities at the company level.

C is incorrect because event-driven strategies typically seek to profit from potential changes in the corporate structure of individual companies. This strategy is considered “bottom up” where the analysis starts at the company level, as opposed to a “top down” approach which starts with macroeconomic analysis.

Introduction to Alternative Investments
LOS d
Section 3.1.1, 3.1.3

- 120** A hedge fund with \$225 million of initial capital charges a management fee of 1% and an incentive fee of 10%. The management fee is based on assets under management at year-end, and the incentive fee is calculated independently from the management fee. Assuming the fund earns a 15% return at year-end, total fees earned by the hedge fund during the year are *closest* to:
- A** \$5.96 million.
 - B** \$5.70 million.
 - C** \$5.63 million.

A is correct. Total fees earned by the hedge fund are closest to \$5.96 million:

$$\begin{aligned}
 \text{Year-end value} &= \$225 \text{ million} \times 1.15 = \$258.75 \text{ million} \\
 \text{Management fee} &= \text{Year-end value} \times \text{Management fee \%} \\
 &= \$258.75 \text{ million} \times 1\% = \$2.5875 \text{ million} \\
 \text{Incentive fee} &= (\text{Year-end value} - \text{Beginning value}) \times \text{Incentive fee \%} \\
 &= (\$258.75 \text{ million} - \$225 \text{ million}) \times 10\% = \\
 &\quad \$3.375 \text{ million} \\
 \text{Total fees} &= \text{Management fee} + \text{Incentive fee} \\
 &= \$2.5875 \text{ million} + \$3.375 \text{ million} = \$5.9625 \text{ million} = \\
 &\quad \$5.96 \text{ million}
 \end{aligned}$$

C is incorrect because \$5.63 million results from calculating the management fee based on assets under management at the beginning of the period, rather than the end of the period:

$$\begin{aligned}
 \text{Year-end value} &= \$225 \text{ million} \times 1.15 = \$258.75 \text{ million} \\
 \text{Management fee} &= \text{Beginning value} \times \text{Management fee \%} \\
 &= \$225 \text{ million} \times 1\% = \$2.25 \text{ million} \\
 \text{Incentive fee} &= (\text{Year-end value} - \text{Beginning value}) \times \text{Incentive fee \%} \\
 &= (\$258.75 \text{ million} - \$225 \text{ million}) \times 10\% = \\
 &\quad \$3.375 \text{ million} \\
 \text{Total fees} &= \text{Management fee} + \text{Incentive fee} \\
 &= \$2.25 \text{ million} + \$3.375 \text{ million} = \$5.625 \text{ million} = \\
 &\quad \$5.63 \text{ million}
 \end{aligned}$$

B is incorrect because \$5.70 million represents the total fees earned if the incentive fee was calculated net of the management fee as opposed to independently from the management fee:

$$\begin{aligned}\text{Year-end value} &= \$225 \text{ million} \times 1.15 = \$258.75 \text{ million} \\ \text{Management fee} &= \text{Year-end value} \times \text{Management fee \%} \\ &= \$258.75 \text{ million} \times 1\% = \$2.5875 \text{ million} \\ \text{Incentive fee} &= (\text{Year-end value} - \text{Beginning value} - \text{Management fee}) \\ &\quad \times \text{Incentive fee \%} \\ &= (\$258.75 \text{ million} - \$225 \text{ million} - \$2.5875 \text{ million}) \times \\ &\quad 10\% = \$3.1163 \text{ million} \\ \text{Total fees} &= \text{Management fee} + \text{Incentive fee} \\ &= \$2.5875 \text{ million} + \$3.1163 \text{ million} = \$5.7038 \text{ million} = \\ &\quad \$5.70 \text{ million}\end{aligned}$$

Introduction to Alternative Investments
LOS e
Section 3.3.1